# Graph Hierarchical Borrowing

Private, Serverless Collaboration for Networks of Small Hospitals

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Most clinical prediction tasks are small at the hospital that needs the model, while the data that would make them large sit in other hospitals that cannot share patient records and are linked, if at all, by a sparse set of bilateral agreements. Federated learning needs a server that every site trusts and fits one model; its private and personalized variants spend the privacy budget over many rounds and must be tuned on data that small hospitals do not have. We propose Graph Hierarchical Borrowing (GHB): each hospital shrinks its own model head toward a leave-one-out estimate of the network, by an amount that its own data decide. In a Gaussian random-effects model we show that every hospital gains almost the same number of "virtual patients", $\sigma^2/(\tau^2 + M)$, so that the smallest hospitals gain the most in relative terms; that differential privacy enters only through the error $M$ of the network estimate; and that computing that estimate by gossip over the agreement graph costs no privacy. For realistic heads, the network makes two joint releases, the sums of every patient's score and information, privatized once for the whole network by secure summation without a server. From them each hospital computes, locally, its leave-one-out prior with the privacy noise propagated, and integrates the unknown heterogeneity out against its own evidence. Nothing is tuned. A lighter variant, GHB-lite, in which each hospital releases on its own, is studied in eighteen simulations; it is more accurate than every baseline tuned with access to the true loss in six of the ten settings of the main comparison, and keeps every hospital close to its own local error. On real data the lighter design failed, and GHB was built from the reasons why. In a pre-registered confirmation on five cancer networks from The Cancer Genome Atlas that no part of its development had seen, GHB was more accurate than every federated and personalized method tuned on twelve development networks, at $\epsilon = 1$ (0.928 of the local error, against 0.976 for the best of them) and at $\epsilon = 10$ (0.915 against 0.919), and at $\epsilon = 1$ it was more accurate than each of them even when tuned on the test folds. At $\epsilon = 10$, such hindsight tuning puts pFedMe ahead (0.898). We report the three pre-registered attempts that preceded GHB, none of which delivered a gain over a well-regularized hospital, and the simulated settings in which GHB-lite is more accurate than GHB.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Applied computing** → **Health informatics**; • **Security and privacy** → **Privacy protections**.

Additional Key Words and Phrases: small data, multi-site learning, decentralized learning, gossip, differential privacy, empirical Bayes, hierarchical models, personalization, data silos, healthcare

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## 1 INTRODUCTION

Health systems produce data at an enormous rate, yet most clinical learning problems are small. Rare diseases are individually rare but collectively common: more than six thousand are known, affecting an estimated 3.5% to 5.9% of the population, and most have cohorts of tens or hundreds of patients worldwide [Nguengang Wakap et al. 2020; Schaefer et al. 2020]. Prediction models in common conditions are routinely built on a few dozen outcome events [Peduzzi et al. 1996; Riley et al. 2019]. During the first months of COVID-19, hundreds of models were developed from small, local samples, and systematic reviews found almost none fit for clinical use [Roberts et al. 2021; Wynants et al. 2020]. Even widely deployed systems turn small at the point of use: a proprietary sepsis model reached an area under the curve of 0.63 on external validation [Wong et al. 2021], and pneumonia classifiers learned to recognize the hospital that produced an image rather than the disease [Zech et al. 2018]. Health data are abundant in aggregate and scarce where they are needed, at a specific site, for a specific outcome. Healthcare is *collectively big and locally small* [Simões et al. 2026].

The data that would close the gap usually exist, but in other hospitals. Each holds a slice of the population under its own legal, ethical and technical constraints [Cohen and Mello 2018; European Parliament and Council of the European Union 2016; Price and Cohen 2019]. Training a model on records from several hospitals is secondary use, which needs its own legal basis and frequently excludes the export of patient-level records; agreements that allow it are typically bilateral, slow to negotiate, and limited to some hospitals and some data [European Parliament and Council of the European Union 2025]. The resulting network is sparse, and in many regions no institution is trusted, mandated or funded to act as its centre. The hospitals with the smallest samples, which need collaboration most, are also those whose data are most exposed by any release, because their few patients are easiest to single out [Rocher et al. 2019].

Four families of methods address parts of this problem, and each gives up something the setting cannot spare. Federated learning keeps data local but needs a server that every site trusts, and its standard form fits one model, which serves the typical site and can be worse than local estimation for atypical or large ones [Hsieh et al. 2020; McMahan et al. 2017]. Personalized federated learning fits a model per site but still needs the server [Collins et al. 2021; Li et al. 2021a]. Decentralized learning removes the server, but its healthcare forms either lack formal privacy or share full models [Kalra et al. 2023; Warnat-Herresthal et al. 2021]. Differentially private versions of all of them spend the privacy budget over many rounds of training, and the noise, calibrated to each site's sensitivity, is largest at the smallest sites, where it erases the benefit of collaboration for exactly the hospitals that need it [Bagdasaryan et al. 2019; McMahan et al. 2018]. Distributed meta-analysis, the classical answer, is exact and auditable, but it estimates a single population model and does not use learned representations or the structure of the network [DerSimonian and Laird 1986; Duan et al. 2020]. No existing method meets the five requirements that the setting imposes at once: data stay local, every release is formally private, no server is needed, the smallest hospitals gain at least as much as the largest, and heterogeneity is modeled rather than assumed away (Section 3).

This paper proposes *Graph Hierarchical Borrowing* (GHB). Its central idea is old: when several units estimate related quantities, each should shrink its own estimate toward what the others suggest, by an amount set by how much the units differ [Efron and Morris 1973; Stein 1956]. What is new is doing this when the network estimate must be private and computed without a server, and characterizing exactly what each constraint costs. The network releases two privatized sums over all of its patients, computed by secure summation and gossip over the agreement graph. Each hospital turns them, locally and at no further privacy cost, into a leave-one-out prior for its own head with the privacy noise propagated, and combines it with its own noise-free data, with a weight

that its own evidence decides. Privacy noise therefore never touches a hospital's own data, only what it borrows, and it is calibrated to the whole network rather than to the smallest hospital.

*Contributions.*

(1) **A formal setting.** We state the problem of a sparse, serverless network of small hospitals as five requirements and show why existing data-exchange agreements and methods do not meet them together (Sections 2 and 3).

(2) **Theory in closed form.** In a Gaussian random-effects model we derive the optimal combination of a local and an imperfect population estimate and show that every hospital gains almost the same number $B = \sigma^2/(\tau^2 + M)$ of virtual patients; that privacy enters only through the error $M$ of the population estimate, with noise divided by the size of the network; and that gossip is free post-processing whose error decays at a rate set by the spectral gap (Section 4).

(3) **A method for realistic model heads.** For likelihood-based heads on top of a public representation, the network releases the privatized sums of its patients' scores and information; each hospital computes its leave-one-out prior from them with the privacy noise propagated, uses the estimates of other networks and a published model when they exist, and integrates the unknown heterogeneity out against its own evidence, so that it borrows strongly when its data agree with the network and falls back to its own data when they conflict. A privacy guarantee covers the whole protocol, and a lighter variant, GHB-lite, needs no secure summation (Section 5).

(4) **An evaluation designed to be hard to pass.** Eighteen simulations test the theory when its assumptions fail, compare GHB-lite and GHB with one-shot and iterative baselines tuned with access to the true loss, attack them with atypical and dishonest hospitals, wrong prior knowledge, failing links and repeated releases, ablate each component, vary each setting, and measure what borrowing does to a protected group of patients (Section 6). On real data, GHB was developed on twelve cancer networks and confirmed once, under pre-registration, on five unseen ones, against six federated and personalized methods tuned as a consortium could and as no consortium can (Section 7).

(5) **Transparent development.** We report every version of GHB, why each was revised, the three pre-registered attempts that did not deliver on real data, the extensions that did not earn a place, and the settings in which the lighter variant is better, with all code, data and every number regenerable (Sections 6 and 7 and the online supplement).

*Main findings.* The theory predicts, and the simulations confirm, that every hospital gains about the same number of virtual patients, $B \approx 46$ in the main simulated network. That multiplies the effective sample size of hospitals with 60 patients or fewer by 2.28 and of those with 300 or more by 1.08. In simulation, GHB-lite is more accurate than every tuned baseline in six of the ten settings of the main comparison, and its worst hospital stays within a factor 1.06 of its own local error in every private setting except the most heterogeneous. DP-FedAvg [McMahan et al. 2018] with fine-tuning, in contrast, leaves some hospital 1.39 to 2.96 times worse off than alone, and the other iterative methods do worse still. In those simulated worlds GHB-lite is also more accurate than GHB (0.0138 against 0.0159 at $\tau^2 = 0.05$), because its inversion is exact there and its releases carry the heterogeneity. On real hospitals the order reverses. GHB-lite was no better than a local model with the same prior on twelve cancer networks. GHB, frozen on those networks and confirmed once on five unseen ones, reduced held-out log-loss to 0.928 of local estimation at $\epsilon = 1$ and 0.915 at $\epsilon = 10$. That was lower than every federated and personalized method tuned on the development networks, the best of which reached 0.976 and 0.919. Tuned on the test folds, which no hospital can do, the six methods remain less accurate than GHB at $\epsilon = 1$, while at $\epsilon = 10$ pFedMe [T. Dinh et al.

2020] leads (0.898). Tuning GHB the same ways changes it little (0.907 and 0.904 at $\epsilon = 10$), so it stays ahead of every method tuned on development networks and of all but pFedMe tuned with hindsight.

*Structure of the paper.* The paper follows the path from the problem to the evidence. Section 2 reviews the four families of methods that address parts of the problem, and ends with what none of them delivers. Section 3 turns that gap into a precise setting, five requirements and a threat model, and measures the existing approaches against them. Section 4 then solves the core of the problem exactly, in a Gaussian random-effects model. It shows what each hospital gains from borrowing, what privacy costs and what the absence of a server costs, and how to handle a heterogeneity that is not known. Section 5 carries this theory to the model heads used in practice. It places borrowing inside an architecture, specifies what the network releases and how each hospital turns it into a prior, describes the lighter variant GHB-lite, and proves the privacy of the whole protocol. The evidence comes in two steps. Section 6 tests the theory and GHB-lite in eighteen simulations in which the truth is known, against tuned baselines, under attack and component by component. Section 7 moves to real hospital networks, where GHB-lite failed and GHB was built, frozen and confirmed under pre-registration. Section 8 draws the consequences: when to use each protocol, what the method and the evidence do not cover, and what remains open. Section 9 concludes. A companion survey [Simões et al. 2026] develops the notion of small data used here and reviews the wider literature on small data in healthcare.

## 2 BACKGROUND AND RELATED WORK

GHB sits at the intersection of four lines of work: distributed statistics, federated learning, decentralized learning and differential privacy. We review each with the question that matters here: what it delivers to a network of small, siloed hospitals without a trusted centre, and what it leaves open. We start with what "small" means, because it fixes what a method must deliver.

### 2.1 Small data and siloed smallness

A dataset is small for a task when the information it carries falls short of what the task needs. The companion survey [Simões et al. 2026] makes this precise. The *effective sample size* $n^{\text{eff}}$ of a dataset is the number of independent, correctly labeled records from the target population that would carry the same information; to first order,

$$n^{\text{eff}} = n \cdot \frac{1}{\text{DEFF}} \cdot \rho_L \cdot (1 - 2\eta)^2 \cdot \rho_S, \tag{1}$$

where the design effect DEFF accounts for repeated records per patient [Kish 1965], $\rho_L$ for missing labels, $\eta$ for label noise [Angluin and Laird 1988] and $\rho_S$ for the shift between the data and the population of use. The *smallness index* $\sigma_{\mathcal{T}} = n^{\star}_{\mathcal{T}}/n^{\text{eff}}$ compares it with the sample complexity $n^{\star}_{\mathcal{T}}$ of the task, for which the criteria of Riley et al. [2019] give closed forms for clinical prediction models; a dataset is small when $\sigma > 1$. A task is *siloed-small* when every one of $K$ custodians holds a dataset that is small for it while their union is not, after accounting for the heterogeneity between custodians. A siloed-small task can be solved without collecting a single new patient; the obstacle is access. Two features of that definition shape what follows. Heterogeneity can make the union much less informative than the sum of its parts, so a method must estimate how much the custodians differ rather than assume they are exchangeable. And the gain from collaboration is naturally measured in effective patients: Section 4 shows that hierarchical borrowing adds a computable number of them to every hospital.

### 2.2 Pooling without moving data: distributed statistics

If small data are the problem, pooling across sites is the obvious remedy. Long before federated learning, biostatistics developed ways to fit models across sites without moving records. Meta-analysis combines site-level estimates, and random-effects meta-analysis explicitly models between-site heterogeneity through a variance $\tau^2$ that can be estimated from the site estimates alone [DerSimonian and Laird 1986]. Individual participant data meta-analysis is more powerful but needs record-level access [Riley et al. 2010]. Distributed regression exchanges sufficient statistics or gradients: GLORE fits logistic regression across sites by exchanging Newton-Raphson summaries [Wu et al. 2012], DataSHIELD wraps such procedures in disclosure controls [Gaye et al. 2014], and one-shot surrogate-likelihood methods such as ODAL obtain near-pooled estimates with one round of communication [Duan et al. 2020; Jordan et al. 2019]. Infrastructures that send analyses to the data rather than data to a centre, such as the OHDSI network on the OMOP common data model [Hripcsak et al. 2015] and the Personal Health Train [Beyan et al. 2020; Deist et al. 2020], have trained models on tens of thousands of patients without moving their records. These methods are exact, auditable and well understood. What they leave open is threefold: they estimate one model for the population, not one per hospital; their disclosure controls are not formal privacy guarantees; and most assume a coordinator that collects the summaries.

Empirical Bayes adds what meta-analysis lacks. Separate estimation of three or more related means is inadmissible under squared error, and shrinkage toward a common mean dominates it [Efron and Morris 1973; Stein 1956]; the hierarchical model behind random-effects meta-analysis yields, for each site, a posterior that combines its own estimate with the population [Gelman et al. 2013]. This is the statistical core of GHB. What has not been worked out is how much it costs to obtain the population estimate privately and without a coordinator, and how to make the combination robust when the heterogeneity itself must be estimated from noisy releases.

### 2.3 Federated and personalized learning

Machine learning took up the same problem a generation later, with models that are trained rather than estimated. Federated averaging [McMahan et al. 2017] trains a shared model by alternating local gradient steps with averaging at a server, and healthcare adopted it quickly [Nguyen et al. 2022; Rieke et al. 2020; Xu et al. 2021]. Federated models have approached the quality of centrally pooled data in brain tumor segmentation [Sheller et al. 2020], improved the prediction of oxygen needs in COVID-19 across 20 institutions [Dayan et al. 2021] and improved tumor delineation across 71 sites [Pati et al. 2022]; FLamby provides cross-silo healthcare benchmarks with natural splits [Ogier du Terrail et al. 2022]. Its central difficulty is heterogeneity: when site distributions differ, local updates drift apart and the averaged model can be worse than local ones [Hsieh et al. 2020; Zhao et al. 2018]. Proximal terms [Li et al. 2020], control variates [Karimireddy et al. 2020], site-specific normalization [Li et al. 2021b] and minimax objectives [Mohri et al. 2019] reduce the drift but still produce one model.

Personalized federated learning stops looking for one model. Multi-task learning [Smith et al. 2017], meta-learning [Fallah et al. 2020], Moreau-envelope regularization [T. Dinh et al. 2020], Ditto [Li et al. 2021a], shared representations with local heads [Collins et al. 2021] and clustered federated learning [Ghosh et al. 2020] all learn a model per site while sharing statistical strength, and the theory of shared representations explains why: a common low-dimensional representation reduces each site's problem to that of a small head [Du et al. 2021; Maurer et al. 2016; Tripuraneni et al. 2020]. GHB adopts this split. What personalized federated learning leaves open is the server, which every site must trust, and the price of personalization: how strongly each site should rely on the others is a tuning parameter, usually set on validation data that small sites do not have.

Fairness in federated learning has two meanings that are easily confused. Fairness across sites asks that no site be badly served by what the network learns; agnostic federated learning [Mohri et al. 2019] and Ditto [Li et al. 2021a] pursue it, and Section 6 measures it for every method as the error of the worst hospital relative to its own local estimate. Fairness across groups of patients asks that a model's errors or decisions not depend on a protected attribute such as sex or ethnicity. FAIR-FATE [Salazar et al. 2023], for example, has the server weigh every client's update by its group fairness and adds a fairness-aware momentum term to the global model. The two can conflict, and they need different machinery: the second requires the protected attribute and a fairness metric, which a method built only for accuracy does not use. GHB is built for accuracy at each hospital. The first kind of fairness is a property it is evaluated on throughout. The second it does not target, and Simulation 18 measures what borrowing does to it, against FAIR-FATE [Salazar et al. 2023] (Section 6); controlling it is open (Section 8.3).

### 2.4 Decentralized learning

Federated and personalized learning both keep a server at the centre of the network. Decentralized SGD replaces the server with neighbour averaging over a communication graph and can match centralized rates when the graph mixes well [Koloskova et al. 2020; Lian et al. 2017; Nedić and Ozdaglar 2009]; gossip algorithms make averaging robust to asynchrony and give rates in terms of the spectral gap of the graph [Boyd et al. 2006; Xiao and Boyd 2004]. Personalized decentralized learning over a similarity graph was introduced by Vanhaesebrouck et al. [2017] and extended with differential privacy by Bellet et al. [2018], and decentralization itself amplifies privacy, because each party sees only what reaches it through the graph [Cyffers and Bellet 2022]. In healthcare, Swarm Learning [Warnat-Herresthal et al. 2021] coordinates peer-to-peer training through a blockchain across clinical use cases, ProxyFL shares differentially private proxy models among peers [Kalra et al. 2023], and peer-to-peer training has been shown for brain and tumor segmentation [Roy et al. 2019; Tedeschini et al. 2022]. These demonstrations are few and mostly simulated. Their methods either do not provide formal privacy by default or share full models, and none models the heterogeneity between hospitals explicitly.

### 2.5 Privacy in multi-site learning

Removing the server, or keeping the data local, does not by itself protect patients. Sharing parameters or gradients is not private by default: gradients can be inverted to reconstruct training images, including in medical federated training [Geiping et al. 2020; Hatamizadeh et al. 2023; Zhu et al. 2019], and trained models leak membership [Shokri et al. 2017]. Differential privacy bounds what any output reveals about an individual [Dwork et al. 2006; Dwork and Roth 2014] and has been adapted to deep learning [Abadi et al. 2016; Mironov 2017] and to federated training [McMahan et al. 2018]; zero-concentrated accounting makes the composition of many releases tight [Bun and Steinke 2016]. Its cost is well documented and falls unevenly: noise is calibrated to sensitivity, which scales inversely with sample size, so private training is least accurate for small datasets, high-dimensional models and under-represented groups [Bagdasaryan et al. 2019; Tramèr and Boneh 2021]. Secure aggregation reveals only sums [Bonawitz et al. 2017], and cryptographic computation protects the process but not what its output reveals. For a network of small hospitals, the consequence is direct: a private method that releases often, or releases high-dimensional objects, or calibrates noise to each site separately without regard to its size, cancels the benefit of collaboration for the hospitals that need it most.

### 2.6 What the literature leaves open

Each line of work supplies a piece. Distributed statistics supplies exact, auditable estimation of a population model; empirical Bayes supplies the optimal way to combine it with each site's own estimate; personalized federated learning supplies the split between a shared representation and local heads; decentralized learning supplies serverless computation with rates set by the graph; and differential privacy supplies a formal guarantee and its accounting. What is missing is their combination for a network of small hospitals: a method that releases little, so that privacy is spent once and at the population level; that personalizes without tuning on data the smallest sites do not have; that runs over whatever agreements exist; and whose benefit to each hospital, and cost to each patient, can be stated in advance. The next section turns these needs into requirements, and Section 4 derives a method that meets them.

## 3 PROBLEM SETTING

The gaps of Section 2.6 become concrete once the setting is stated precisely. This section states it, turns the needs of a network of small hospitals into five requirements, fixes the threat model, and measures the approaches of Section 2 against those requirements.

There are $K$ hospitals, the vertices of an undirected connected graph $\mathcal{G} = (V, E)$. An edge means that two hospitals have an agreement allowing them to exchange privatized model summaries; $\mathcal{G}$ is sparse, with $|E| = O(K)$, and there is no vertex that every hospital trusts. Hospital $k$ holds $n_k$ labeled records from its own population $P_k$ and wants a model that performs well on that population. In the terms of Section 2.1, the task is siloed-small, and the obstacle is custodial. We require:

(R1) no patient-level record leaves its hospital;
(R2) every released quantity satisfies $(\epsilon, \delta)$-differential privacy with respect to the patients of the releasing hospital;
(R3) no central server: computation uses only messages along edges of $\mathcal{G}$;
(R4) the smallest hospitals benefit at least as much as the largest, in relative terms;
(R5) heterogeneity across hospitals is modeled rather than assumed away.

Requirements (R4) and (R5) exclude a single global model: a global model serves the typical site, not the smallest or most atypical, and the private cost of fitting it falls on the smallest participants (Section 2.5).

*Why existing data-exchange agreements are not enough.* Hospitals already exchange data under protocols, but those protocols do not remove the problem. Most of them serve primary use: a patient's record follows the patient to the next point of care. Training a model on the records of many hospitals is secondary use, which needs its own legal basis, ethics approval and data-access permit, and frequently excludes the export of patient-level records altogether [European Parliament and Council of the European Union 2016, 2025]. Where agreements for secondary use exist, they are typically bilateral, slow to negotiate and limited to a subset of hospitals and data, so the network they form is the sparse graph $\mathcal{G}$ assumed here: GHB is designed to run over existing agreements, not to replace them. Even where pooling is lawful, a central copy of identifiable records is not the least invasive option, given how easily such records are re-identified [Rocher et al. 2019], and data minimization favours releasing a few privatized, low-dimensional sums over the network. Existing infrastructures that move analyses to the data rather than data to a centre, such as the OHDSI network [Hripcsak et al. 2015], DataSHIELD [Gaye et al. 2014] and the Personal Health Train [Beyan et al. 2020; Deist et al. 2020], are complementary: they provide the channels, and GHB specifies what to compute through them, how to combine the results and what it costs in privacy. Finally, even unrestricted pooling yields one model for all hospitals, which Section 6 shows

to be worse than local estimation for atypical and large hospitals; the personalization of GHB is useful whether or not privacy binds. The converse also holds. When a network already has a trusted environment in which pooling is lawful, such as a national registry or a secure processing environment under the European Health Data Space, requirement (R3) is moot, and the hierarchical borrowing of Section 4.1 can run inside that environment without gossip or privatized releases.

*Threat model.* GHB protects patients against honest-but-curious hospitals and against any observer of the messages exchanged along $\mathcal{G}$, under a differential-privacy guarantee on everything that leaves a hospital. Its joint releases are computed by secure summation, which assumes that the hospitals do not collude to remove their noise shares (Proposition 5.1 states what collusion costs); its lighter variant, GHB-lite, needs no such assumption, because each hospital's release is private on its own. Hospitals are assumed to follow the protocol. A hospital that lies about its contribution is not excluded by the protocol itself, and Section 6.4 tests what such lies do and how screening contains them. Site sizes $n_k$ are treated as public; Sections 4.2 and 5.4 list every assumption under which the privacy guarantees hold.

*How existing approaches fare.* Table 1 compares the approaches of Section 2 with the requirements. Centralized pooling violates (R1). Federated averaging [McMahan et al. 2017] violates (R3) and, under heterogeneity, (R4) and (R5). Personalized federated learning satisfies (R4) and (R5) but still needs a server. Existing decentralized approaches satisfy (R3); Swarm Learning [Warnat-Herresthal et al. 2021] does not provide formal differential privacy by default, while ProxyFL [Kalra et al. 2023] does but shares full proxy models. Synthetic data sharing satisfies (R1) and (R3) only at the cost of utility or privacy [Jordon et al. 2019; Stadler et al. 2022]. Distributed meta-analysis is the closest classical relative and satisfies most requirements, but does not exploit learned representations or graph structure. The comparison is one of design properties, not of measured performance: the GHB row records what holds by construction under the assumptions of Propositions 4.1, 4.3 and 5.1 below, while the other rows describe each method in its standard form, and most of them can be extended to satisfy more requirements. Measured performance is the subject of Section 6.

## 4 HIERARCHICAL BORROWING IN A MODEL THAT CAN BE SOLVED EXACTLY

We first analyze the core of the problem in the simplest model that keeps its structure: one parameter per hospital, a Gaussian random-effects model, and a population estimate that must be obtained privately and without a server. This yields closed forms for what each hospital gains (Section 4.1), what privacy costs (Section 4.2) and what the absence of a server costs (Section 4.3). Sections 4.4 and 4.5 then remove two simplifications, known heterogeneity and exchangeable hospitals, and Section 5 carries the results to the model heads used in practice.

### 4.1 Borrowing with an imperfect population estimate

Each hospital has a parameter $\theta_k \in \mathbb{R}$ (one coordinate of a model head; the vector case is coordinatewise) and an asymptotically normal local estimate:

$$\theta_k = \mu + u_k, \quad u_k \sim \mathcal{N}(0, \tau^2); \qquad \hat{\theta}_k = \theta_k + e_k, \quad e_k \sim \mathcal{N}(0, v_k), \quad v_k = \sigma^2/n_k, \tag{2}$$

with all $u_k, e_k$ independent. Here $\tau^2$ is the between-hospital heterogeneity, the quantity that random-effects meta-analysis estimates [DerSimonian and Laird 1986], and $\sigma^2$ absorbs the inverse Fisher information per patient. This is the classical setting of empirical Bayes [Efron and Morris 1973; Gelman et al. 2013]. What is new here is that the population mean $\mu$ must be estimated privately and without a server, so hospital $k$ only has access to some estimate $\tilde{\mu}$ with error.

Table 1. Approaches for sparse, siloed hospital networks against requirements (R1) to (R5), as design properties of each method in its standard form. ✓ satisfied; ○ partially or optionally; ✗ not satisfied. Entries for GHB hold by construction under the assumptions of Sections 4 and 5.4; (R2) and (R3) assume secure summation by hospitals that do not collude (Proposition 5.1).

| Approach | (R1) local data | (R2) formal DP | (R3) no server | (R4) small sites gain | (R5) heterogeneity |
|---|---|---|---|---|---|
| Centralized pooling | ✗ | ✗ | ✗ | ○ | ○ |
| Federated averaging [McMahan et al. 2017] | ✓ | ○ | ✗ | ✗ | ✗ |
| Personalized FL [Collins et al. 2021; Li et al. 2021a] | ✓ | ○ | ✗ | ✓ | ✓ |
| Swarm Learning [Warnat-Herresthal et al. 2021] | ✓ | ✗ | ✓ | ○ | ✗ |
| ProxyFL [Kalra et al. 2023] | ✓ | ✓ | ✓ | ○ | ○ |
| Synthetic data sharing [Jordon et al. 2019] | ✓ | ○ | ✓ | ○ | ✗ |
| Distributed meta-analysis [DerSimonian and Laird 1986; Duan et al. 2020] | ✓ | ○ | ○ | ✓ | ✓ |
| **GHB (this paper)** | ✓ | ✓ | ✓ | ✓ | ✓ |

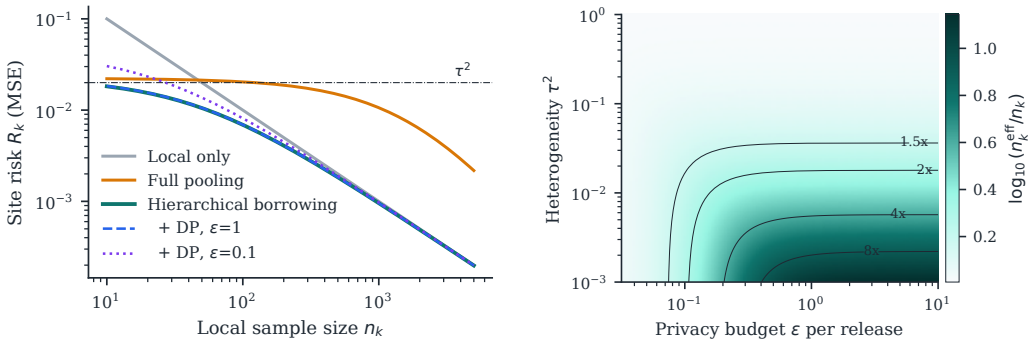



Fig. 1. Left: risk of hospital $k$ as a function of its size $n_k$, with $K = 12$, other hospitals of 200 patients, $\sigma^2 = 1$, $\tau^2 = 0.02$ and $\delta = 10^{-5}$. Full pooling is excellent for small sites until it hits the heterogeneity floor $\tau^2$, and then harms large ones; hierarchical borrowing (Proposition 4.1), with $\tau^2$ and $M$ known, is never worse than either in expected risk. Privacy at $\epsilon = 1$ is almost free; at $\epsilon = 0.1$ it reduces but does not remove the gain. Right: relative gain in effective sample size $n_k^{\text{eff}}/n_k$ for a hospital with $n_k = 50$, as a function of heterogeneity and privacy budget, from Equations (3) and (4).

Proposition 4.1 (Borrowing with an imperfect prior). *Let $\tilde{\mu}$ be any estimate of $\mu$ that is independent of $(u_k, e_k)$ and has mean squared error $M = \mathbb{E}(\tilde{\mu} - \mu)^2$. Among estimators $\tilde{\theta}_k = \alpha\hat{\theta}_k + (1-\alpha)\tilde{\mu}$, the risk $R_k(\alpha) = \mathbb{E}(\tilde{\theta}_k - \theta_k)^2$ is minimized at*

$$\alpha_k^\star = \frac{\tau^2 + M}{\tau^2 + M + v_k}, \qquad R_k^\star = \frac{v_k\,(\tau^2 + M)}{v_k + \tau^2 + M} \;\le\; \min\{v_k,\ \tau^2 + M\}.$$

*Equivalently, the effective sample size of hospital $k$ becomes*

$$n_k^{\text{eff}} = \frac{\sigma^2}{R_k^\star} = n_k + \underbrace{\frac{\sigma^2}{\tau^2 + M}}_{\text{borrowed patients } B}. \tag{3}$$

Proof. $\tilde{\theta}_k - \theta_k = \alpha e_k + (1-\alpha)(\tilde{\mu} - \mu) - (1-\alpha)u_k$. The three terms are mutually independent, and the first and last have zero mean, so the cross terms vanish; the middle term has second moment $M$. Hence $R_k(\alpha) = \alpha^2 v_k + (1-\alpha)^2(\tau^2 + M)$. This is a convex quadratic in $\alpha$, minimized at $\alpha_k^\star$; substituting gives $R_k^\star$. The bound follows because $R_k^\star$ is half the harmonic mean of $v_k$ and $\tau^2 + M$. Finally $\sigma^2/R_k^\star = \sigma^2/v_k + \sigma^2/(\tau^2 + M) = n_k + B$. □

Two remarks fix the scope of Proposition 4.1. First, the proof uses only first and second moments and independence, so it does not need the normality of Equation (2); normality matters only for reading $R_k^\star$ as a posterior variance. Second, $R_k$ is an expectation over the site effect $u_k$ as well as over the data: it is the risk of a hospital drawn from the population of hospitals, not of a particular one. For an unbiased $\tilde{\mu}$ and a given $u_k$, the risk is $\alpha^2 v_k + (1-\alpha)^2(M + u_k^2)$, which exceeds $v_k$ when $|u_k|$ is large compared with $\tau$. A genuinely atypical hospital can therefore lose by borrowing. This is why $\tau^2$ is estimated from the data rather than assumed small, why Section 4.4 lets each hospital's own data temper how much it borrows, and why the structured prior of Section 4.5 lets similarity decide which hospitals borrow from which.

Equation (3) is the central quantity of the method. Collaboration gives every hospital almost the same number $B$ of *virtual patients*, regardless of its size; $B$ depends on $k$ only through the leave-one-out error $M$, which varies little across hospitals when there are many of them. A hospital with 30 patients and one with 3,000 both gain $B$; the first may double or triple its effective sample, the second barely notices. Requirement (R4) is therefore satisfied by construction. $B$ is limited by heterogeneity: as $\tau^2 \to \infty$, $B \to 0$ and each hospital should act alone; as $\tau^2 \to 0$ with $M \to 0$, $B \to \infty$ and full pooling is optimal. Hierarchical borrowing interpolates between the two, using the data to decide where. With $\tau^2$ and $M$ known it never does worse than local estimation, which is not true of a global model (Figure 1); when both are estimated from the privatized releases the loss is small in simulation, but no longer zero (Section 6).

### 4.2 The cost of privacy

Proposition 4.1 holds for any estimate $\tilde{\mu}$. Requirements (R2) and (R3) constrain how that estimate can be obtained, and the next two results quantify what each constraint costs. Hospital $k$ needs $\tilde{\mu}$ from the others, and each release must be private (R2). Note what does not need privatizing: hospital $k$ uses its own estimate $\hat{\theta}_k$ without noise, since that never leaves the hospital. Privacy affects only $M$.

Proposition 4.2 (Privatized population estimate). *Suppose $\hat{\theta}_j$ is the mean of per-patient contributions clipped to $[-C, C]$, so that replacing one patient changes $\hat{\theta}_j$ by at most $\Delta_j = 2C/n_j$, and that clipping introduces no bias. Each hospital $j$ releases $\hat{\theta}_j + \xi_j$ with $\xi_j \sim \mathcal{N}(0, s_j^2)$ and $s_j = \Delta_j\sqrt{2\ln(1.25/\delta)}/\epsilon$, which is $(\epsilon, \delta)$-differentially private for $\epsilon \in (0,1)$ by the Gaussian mechanism [Dwork and Roth 2014]. For the leave-one-out average $\tilde{\mu} = \frac{1}{K-1}\sum_{j\neq k}(\hat{\theta}_j + \xi_j)$,*

$$M = \frac{1}{(K-1)^2}\sum_{j\neq k}\left(\tau^2 + \frac{\sigma^2}{n_j} + \frac{8C^2\ln(1.25/\delta)}{n_j^2\,\epsilon^2}\right). \tag{4}$$

Proof. The terms $u_j + e_j + \xi_j$ are independent with zero mean and variances $\tau^2 + \sigma^2/n_j + s_j^2$, and $s_j^2 = 4C^2 \cdot 2\ln(1.25/\delta)/(n_j^2\epsilon^2)$. Independence from $(u_k, e_k)$ holds because $k$ is excluded. □

Three consequences follow. First, privacy noise is divided by $(K-1)$, so it vanishes as the network grows, while the local estimate remains noise-free. Second, the privacy term scales as $n_j^{-2}$ while the sampling term scales as $n_j^{-1}$: privacy is cheap for large contributors and expensive for tiny ones, which is the formal version of the uneven cost of private learning documented in Section 2.5. The remedy is to let small hospitals release jointly: with secure aggregation [Bonawitz et al. 2017] among neighbours, noise can be calibrated to the sensitivity of the sum, which is set by the total size of the group rather than by each member. Third, inverse-variance weighting of the releases, rather than equal weights, reduces $M$ further and is what a practical implementation should use; Equation (4) is the simplest case.

*Scope of the privacy guarantee.* Five conditions delimit what Proposition 4.2 guarantees. (a) Neighbouring datasets differ in one patient's record at one hospital. A patient whose records are held by $m$ hospitals appears in $m$ releases, and by composition is protected at level $(m\epsilon, m\delta)$. (b) The site sizes $n_j$ and the per-patient variance $\sigma^2$ are treated as public; if they are not, they must be released privately as well. (c) A head $\theta_k \in \mathbb{R}^p$ must be released as one vector, with per-patient contributions clipped in $\ell_2$ norm to $C$ and isotropic noise calibrated to the $\ell_2$ sensitivity $2C/n_j$, which gives $(\epsilon, \delta)$ for the whole vector; calibrating each coordinate separately would compose to $(p\epsilon, p\delta)$. (d) For estimators that are not means of clipped contributions, $\Delta_j$ is replaced by the sensitivity of the released statistic. For a head fitted by maximum likelihood, the simplest choice is the mean score at a public point $c$, $\frac{1}{n_j}\sum_i \nabla\ell_i(c)$, whose sensitivity is $2L/n_j$ for an $L$-Lipschitz loss ($L = \max\|x\|$ for logistic regression). The parameter is then recovered by public post-processing, by matching the score's expectation under the public feature distribution (Section 5.3); this is exact when a hospital's features follow that distribution and approximate under case-mix shift. Evaluated at the anchor, $c = \mu_0$, the score acts as a control variate: a wrong anchor makes the release noisier, not biased. A regularized minimizer can be released instead, with sensitivity $2L/(n_j\lambda)$ for a $\lambda$-strongly convex objective [Chaudhuri et al. 2011], but it is noisier, and undoing the regularization to first order introduces a bias that grows with the distance between the parameter and the centre of the regularizer (Section 6). (e) Each refit on the same patients is a new release, and the guarantees compose; zero-concentrated accounting [Bun and Steinke 2016], under which the Gaussian mechanism with noise $s$ costs $\rho = \Delta^2/(2s^2)$ and costs add, is far tighter than basic composition over many refits (Section 6).

### 4.3 Computation on a sparse graph without a server

The leave-one-out average needs every hospital's release. Without a server, the hospitals compute it by gossip over $\mathcal{G}$ [Boyd et al. 2006; Xiao and Boyd 2004]. Let $W$ be a symmetric, doubly stochastic matrix supported on the edges of $\mathcal{G}$, for instance Metropolis-Hastings weights $W_{kl} = 1/(1 + \max(d_k, d_l))$ for $(k,l) \in E$, where $d_k$ is the degree of $k$. Starting from the privatized releases $x_j^0 = \hat{\theta}_j + \xi_j$, each round sets $x^{t+1} = Wx^t$.

Proposition 4.3 (Free post-processing and convergence). *(i) Every message exchanged during gossip is a function of the privatized releases only, so the whole gossip phase satisfies the same $(\epsilon, \delta)$ guarantee as the initial release, with no additional privacy cost. (ii) Let $\lambda = \max\{|\lambda_2(W)|, |\lambda_K(W)|\}$. If $\mathcal{G}$ is connected and $W$ has a positive diagonal, then $\lambda < 1$ and*

$$\left\|x^t - \bar{x}\mathbf{1}\right\|_2 \;\le\; \lambda^t \left\|x^0 - \bar{x}\mathbf{1}\right\|_2,$$

*so that the estimate held by any hospital after $t$ rounds has root mean squared error at most $\sqrt{M_{\text{all}}} + \lambda^t \sqrt{\mathbb{E}\|x^0 - \bar{x}\mathbf{1}\|_2^2}$ as an estimate of $\mu$, where $M_{\text{all}} = K^{-2}\sum_j(\tau^2 + \sigma^2/n_j + s_j^2)$ is the mean squared error of the full average. (iii) Hospital $k$ obtains its leave-one-out estimate as $\tilde{\mu}_k^t = (Kx_k^t - x_k^0)/(K-1)$. It converges to the leave-one-out average of Proposition 4.2, and differs from it by $K(x_k^t - \bar{x})/(K-1)$, whose root mean square is at most $\frac{K}{K-1}\lambda^t\sqrt{\mathbb{E}\|x^0 - \bar{x}\mathbf{1}\|_2^2}$.*

Proof. (i) is the post-processing property of differential privacy [Dwork and Roth 2014]: the gossip iterates are a deterministic function of the released values, and each hospital's subsequent computations on its own data are never released. (ii) Since $W$ is symmetric and doubly stochastic, $\mathbf{1}$ is an eigenvector with eigenvalue 1 and $x^t - \bar{x}\mathbf{1} = W^t(x^0 - \bar{x}\mathbf{1})$ lies in the orthogonal complement of $\mathbf{1}$, on which $\|W\|_2 = \lambda$. Connectivity and a positive diagonal make $W$ irreducible and aperiodic, so $\lambda < 1$ [Xiao and Boyd 2004]. The error bound follows from the triangle inequality in $L^2$, since each coordinate deviation is bounded by the norm of the whole vector. (iii) Since $\sum_{j\neq k} x_j^0 = K\bar{x} - x_k^0$, the leave-one-out average is $(K\bar{x} - x_k^0)/(K-1)$, and subtracting it from $\tilde{\mu}_k^t$ leaves $K(x_k^t - \bar{x})/(K-1)$, which is bounded as in (ii). □

Part (iii) also bounds the one departure from Proposition 4.1 that gossip introduces: after finitely many rounds, $\tilde{\mu}_k^t$ still contains a fraction of hospital $k$'s own release, so it is not exactly independent of $(u_k, e_k)$. The residual dependence decays as $\lambda^t$, and a hospital should use $\tilde{\mu}_k^t$ only once $\lambda^t$ is small; Section 6 shows what happens otherwise.

The number of rounds needed for accuracy $\gamma$ is about $\log(1/\gamma)/(1-\lambda)$, so the spectral gap $1-\lambda$ of the collaboration graph determines the communication cost. The spectral gap is related to how well the graph is connected through Cheeger-type inequalities [Chung 1997]. Figure 2 shows this for 24 hospitals under four realistic topologies. A chain of bilateral agreements arranged in a ring mixes slowly; dense regional clusters joined by single bridges are not much better, because the bridges are bottlenecks; a few additional random or two-hop links triple the spectral gap, which after a hundred rounds lowers the consensus error by orders of magnitude. The practical lesson is that *which* agreements a network signs matters more than *how many*: adding a handful of long-range edges is worth more than densifying regions. Privacy amplification by decentralization [Cyffers and Bellet 2022] gives a further benefit on sparse graphs, because each hospital observes only what reaches it through its neighbours.

### 4.4 Unknown heterogeneity

The three propositions take $\tau^2$ as known. It is not, and how a hospital handles that uncertainty decides whether borrowing helps it or harms it.

*Remark 4.4 (Estimating $\tau^2$, or integrating it out).* $\alpha_k^\star$ depends on $\tau^2$, which is itself unknown. Method-of-moments estimators from random-effects meta-analysis [DerSimonian and Laird 1986] need only the released $\hat{\theta}_j + \xi_j$ and their variances $\sigma^2/n_j + s_j^2$, which are public, so they inherit the privacy guarantee by post-processing. With few hospitals, $\tau^2$ is poorly estimated and a weakly informative prior on it is advisable [Gelman et al. 2013]. A plug-in estimate can be avoided altogether. Hospital $k$ can integrate $\tau$ out: summarize the posterior of $\tau$ given the releases (under a half-Cauchy prior; the releases are obtained by flooding, which is post-processing) by $Q$ equally weighted quantiles $\tau_q$, and average its estimates under the priors $\mathcal{N}(\tilde{\mu}_k(\tau_q), \tau_q^2 + M_k(\tau_q))$, weighting each by the marginal likelihood of its own data. This is the posterior mean of $\theta_k$ when $\tau$ is unknown. In the Gaussian model each weight is the density $\mathcal{N}(\hat{\theta}_k; \tilde{\mu}_k(\tau_q), \tau_q^2 + M_k(\tau_q) + v_k)$, so it is closed form; for other heads a Laplace approximation suffices. A hospital whose data agree with the network borrows strongly; one whose data conflict with it puts its weight on wide components and borrows

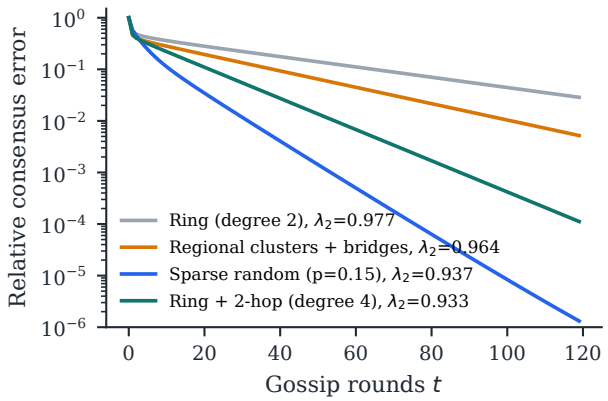



Fig. 2. Gossip consensus error for 24 hospitals on four sparse topologies with Metropolis-Hastings weights. Rates follow the second largest eigenvalue modulus $\lambda$ of the gossip matrix (Proposition 4.3).

little, a smooth version of a rule that keeps the local estimate when it conflicts with the network, which GHB-lite offers as an option (Section 6.4). It costs no privacy, because it uses the hospital's own data only inside the hospital.

Integrating $\tau$ out matters most where a plug-in estimate is least reliable. When releases are noisy, the posterior of $\tau$ is wide, and its mean is driven by the tail of the prior rather than by the data; a plug-in posterior mean then overstates the heterogeneity and switches borrowing off, while a plug-in median or a moment estimate can understate it and make hospitals borrow too much. Averaging over the posterior, weighted by each hospital's own evidence, avoids committing to either. Simulation 16 (Section 6.5) quantifies the difference: with noisy releases, it is one of the largest improvements in GHB-lite.

### 4.5 Structured heterogeneity

The exchangeable model (2) treats all other hospitals as equally relevant. When similarity is structured, for instance by region or case mix, the prior can follow a similarity graph with weights $a_{kl} \geq 0$, which may differ from the communication graph. A Gaussian Markov random field prior [Rue and Held 2005] with graph Laplacian $L$ gives the maximum a posteriori problem

$$\min_{\theta} \sum_{k} \frac{(\theta_k - \hat{\theta}_k)^2}{2v_k} + \frac{\beta}{2} \sum_{k<l} a_{kl}(\theta_k - \theta_l)^2 + \frac{\gamma}{2} \sum_{k} (\theta_k - \mu_0)^2, \tag{5}$$

whose solution satisfies the linear system $(V^{-1} + \beta L + \gamma I)\boldsymbol{\theta} = V^{-1}\hat{\boldsymbol{\theta}} + \gamma\mu_0\mathbf{1}$ with $V = \mathrm{diag}(v_k)$. The Jacobi iteration

$$\theta_k \leftarrow \frac{\hat{\theta}_k/v_k + \beta \sum_l a_{kl}\theta_l + \gamma\mu_0}{1/v_k + \beta d_k + \gamma}, \qquad d_k = \textstyle\sum_l a_{kl},$$

uses only neighbour values and converges for any starting point, because the system matrix is strictly diagonally dominant. This generalizes the personalized peer-to-peer objective of Vanhaesebrouck et al. [2017] with an explicit anchor $\mu_0$, which can come from published knowledge. To respect (R2), neighbours exchange smoothed values computed from privatized releases, and each hospital applies the final update with its own noise-free $\hat{\theta}_k$ locally without releasing it, so Proposition 4.3(i) applies.

## 5 GHB FOR MODEL HEADS

The analysis of Section 4 applies directly to parametric models, and in particular to the last layer of a network on top of a fixed representation. This section builds on that observation. It places hierarchical borrowing inside an architecture in which each source of information is used where it is strongest (Section 5.1), specifies what the network releases and how each hospital turns it into a prior for its own head (Section 5.2), describes the lighter variant GHB-lite, in which each hospital releases on its own (Section 5.3), and assembles the protocol and its privacy guarantee (Section 5.4).

### 5.1 Architecture

GHB keeps the privatized, shared part of the model small and surrounds it with components that cost no privacy (Figure 3). In the terms of the companion survey, each layer draws on a different source, or *provenance channel*, of the information that a small hospital lacks [Simões et al. 2026].

**L0 Public representation (proxy).** A foundation encoder $f_{\phi_0}$ pretrained on public or licensed data for the modality [Moor et al. 2023; Rasmy et al. 2021; Zhou et al. 2023]. It is available to all hospitals without any privacy cost.

**L1 Local self-supervised adaptation (unlabeled data).** Each hospital adapts the encoder to its own unlabeled data with low-rank adapters [Hu et al. 2022] and a self-supervised objective [Chen et al. 2020; He et al. 2022]. Adapters may be averaged by privatized gossip, but need not be.

**L2 Hierarchical heads (pooling).** Each hospital fits a low-dimensional head $\theta_k \in \mathbb{R}^p$ on the adapted features and borrows strength by Propositions 4.1, 4.2 and 4.3, with $\tau$ integrated out (Remark 4.4), or by Equation (5).

**L3 Knowledge anchor (prior knowledge).** The hyperprior mean $\mu_0$ comes from published models or clinical knowledge, so that even a hospital with no neighbours starts from a sensible place.

**L4 Site calibration.** Each hospital calibrates its predictions by conformal prediction on local held-out data [Angelopoulos and Bates 2023; Shafer and Vovk 2008], or by federated conformal prediction when the local calibration set is too small [Lu et al. 2023].

**L5 Targeted procurement.** Posterior variances from L2 identify which patients or subgroups would be most informative to label next [Settles 2009].

Synthetic data is deliberately absent from the training path. It is useful for testing pipelines and for teaching, but a generator fitted to a hospital's data cannot add information about the target that was not already in those data [Simões et al. 2026], and its privacy case is weak [Stadler et al. 2022].

Two features of the architecture deserve emphasis. Keeping the privatized component low-dimensional is not a convenience but a necessity: the norm of Gaussian-mechanism noise grows with $\sqrt{p}$, and private learning is far more accurate on good fixed features than end to end [Tramèr and Boneh 2021]. And the split between a public representation and a private head mirrors the theory of shared representations, in which a common representation reduces the per-task sample complexity to that of the head [Collins et al. 2021; Du et al. 2021; Maurer et al. 2016; Tripuraneni et al. 2020].

### 5.2 What the network releases

Proposition 4.2 assumes that each hospital releases its own clipped mean, so that every hospital's release carries noise calibrated to its own size. Its second consequence points to the remedy: noise calibrated to a *sum* is set by one patient, not by the smallest hospital. GHB takes that remedy to the whole network. The network makes two joint releases, both sums over every patient of every

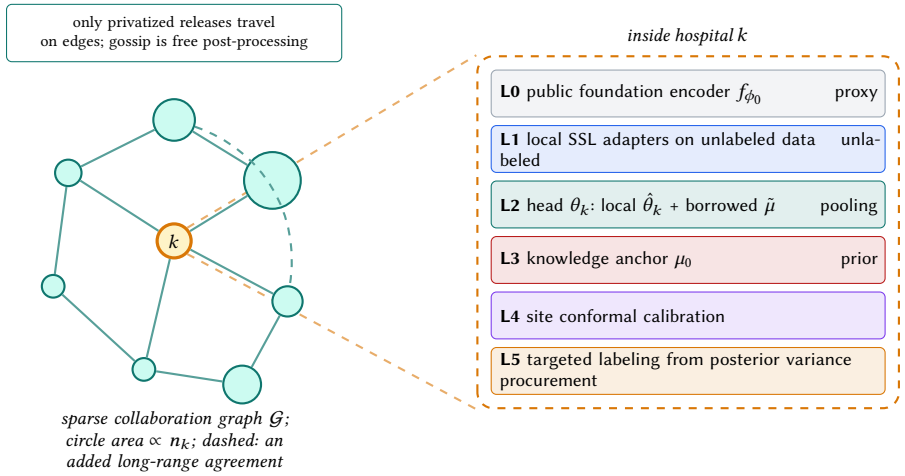



Fig. 3. Graph Hierarchical Borrowing (GHB). Left: hospitals of very different sizes connected by a sparse set of agreements, without a central server. Right: the model held by each hospital, with the source of information used at each layer. Only the low-dimensional head participates in privatized exchange.

hospital, revealed only as totals. Everything a hospital borrows is computed from these two totals and from public information. A hospital uses its own data only inside its walls.

*Two joint releases.* Let $G = \max \|x\|$ bound the features of a likelihood-based head; for logistic regression the per-patient score at the origin is $\psi_i = x_i(y_i - \frac{1}{2})$, and the per-patient information there is $\phi_i = x_i x_i^\top/4$. The network releases

$$\tilde{S} = \sum_k \sum_{i\in k} \psi_i + \xi, \quad \xi \sim \mathcal{N}(0, s_S^2 I_p), \qquad \tilde{J} = \sum_k \sum_{i\in k} \phi_i + E, \quad E_{ab} = E_{ba} \sim \mathcal{N}(0, s_J^2)\ (a \le b).$$

Replacing one patient moves $\sum \psi_i$ by at most $2G$ in $\ell_2$ norm (the bound of scope (d) of Section 4.2, which holds at any evaluation point and is conservative by a factor of two at the origin), and moves the upper triangle of $\sum \phi_i$ by at most $\|\phi_i - \phi_i'\|_F \le G^2/2$. Each release receives half of a zero-concentrated budget $\rho$, so that $s_S = 2G/\sqrt{\rho}$ and $s_J = (G^2/2)/\sqrt{\rho}$, with $\rho + 2\sqrt{\rho \ln(1/\delta)} = \epsilon$ [Bun and Steinke 2016]. The sums are computed by secure summation [Bonawitz et al. 2017]: neighbours agree on pairwise random masks that cancel in the total, every hospital adds its masked contribution and a share $\mathcal{N}(0, s^2/K)$ of the noise, and gossip over $\mathcal{G}$ averages the masked values (Proposition 4.3), so no server is needed and no hospital's own contribution is ever revealed.

*From the releases to the network.* With $A = (\tilde{J}^+ + I/s_0^2)^{-1}$, where $\tilde{J}^+$ is $\tilde{J}$ projected onto the positive semidefinite cone and $s_0 = 2.5$ is the scale of the weakly informative default prior for logistic coefficients [Gelman et al. 2008], the network estimate is one Newton step of the pooled model from the origin,

$$\hat{w} = A\tilde{S}, \qquad V_{\mathrm{dp}} = s_S^2 AA^\top,$$

and $V_{\mathrm{dp}}$, the covariance that the privacy noise adds to $\hat{w}$, is public. In the terms of Proposition 4.2, the privacy part of $M$ is now of order $s_S^2/N^2$ per unit of information, with $N = \sum_k n_k$, rather than an average of terms of order $1/n_j^2$: the smallest hospital no longer sets the noise. Two further sources of context are used when they exist, both by post-processing. When several networks fit the same head, for instance one network per cancer type, each network's $\hat{w}$ is shrunk, coefficient by coefficient,

toward the precision-weighted mean of the other networks' estimates, with a between-network variance taken at the median of its posterior under a half-Cauchy prior; the networks exchange only their $\hat{w}$ and its covariance. When a published model $\mu_0$ with stated uncertainty $\sigma_0^2$ exists, it is combined with $\hat{w}$ by precision weighting unless it conflicts with it ($|\mu_0 - \hat{w}| > 3\sqrt{\sigma_0^2 + V}$ in some coordinate).

*From the network to each hospital.* Hospital $k$ computes its exact score at the network estimate, $s_k = \sum_{i \in k} x_i(y_i - \sigma(x_i^\top \hat{w}))$, and its information $J_k = \sum_{i \in k} x_i x_i^\top / 4$, and removes its own data from the network estimate by one Newton step,

$$\mu_{-k} = \hat{w} - H_{-k}^{-1} s_k, \qquad V_{-k} = H_{-k}^{-1} + V_{\mathrm{dp}}, \qquad H_{-k} = (\tilde{J} - J_k)^+ + I/s_0^2,$$

the leave-one-out estimate that Proposition 4.1 requires, now with its full covariance: sampling error and privacy noise. None of this is released. Hospital $k$ then integrates the unknown heterogeneity out as in Remark 4.4: for each of $Q = 12$ quantiles $\tau_q$ of a half-Cauchy prior with scale 0.5 and each scale $s \in \{0.5, 1, 2.5\}$ of the default prior, it forms the Gaussian prior $\mathcal{N}(\mu_{-k}, \mathrm{diag}(\tau_q^2) + V_{-k})$ multiplied by $\mathcal{N}(0, s^2 I)$, adds three components without borrowing, $\mathcal{N}(0, s^2 I)$, with a quarter of the prior weight between them, fits its head under every component by maximum a posteriori, and averages the fits weighted by the Laplace approximation of the marginal likelihood of its own data. A hospital whose data agree with the network puts its weight on narrow components and borrows strongly. One whose data conflict with the network puts it on wide or non-borrowing components, and so falls back to its own data. No setting is tuned on outcome data: the release split, the half-Cauchy scale, the default scales and the weight of the non-borrowing components are fixed in advance and were frozen before the confirmation of Section 7.

*Why the releases are sums.* The simpler design of GHB-lite below failed on real data, for two reasons that shaped the one above (Section 7). In GHB-lite each hospital releases its own score, and the network maps it back to the parameter scale under a public feature distribution. Real hospitals' case mixes differ from any such distribution, so the mapping is biased. And the per-hospital noise, calibrated to $1/n_k$, is so large at strong privacy that the network estimate carries almost no information. Joint releases remove both problems: the information matrix comes from the hospitals' real data, and the noise is calibrated to the whole network. The price is a stronger trust model for the summation step, which Proposition 5.1 states.

### 5.3 GHB-lite: one release per hospital

Where secure summation is not available, GHB-lite keeps the original design: each hospital releases its mean score at the published anchor,

$$g_k = \frac{1}{n_k} \sum_{i \in k} x_i \big(y_i - \sigma(x_i^\top \mu_0)\big) + \xi_k, \qquad \xi_k \sim \mathcal{N}(0, s_k^2 I_p), \quad s_k = \frac{2G}{n_k \sqrt{2\rho}},$$

which is private on its own, whatever the other hospitals do. Under the public feature distribution, $\mathbb{E} g_k = m(\theta_k) - m(\mu_0)$ with $m(\theta) = \mathbb{E}[x\, \sigma(x^\top \theta)]$, so the network recovers $\tilde{\theta}_k = a_k + I(a_k)^{-1}(g_k + m(\mu_0) - m(a_k))$ by moment matching, linearized first at the pooled moment estimate and then at each hospital's leave-one-out estimate, where $I$ is the Fisher information under the public feature distribution. Each hospital forms the posterior of $\tau$ from the recovered releases, takes $Q$ quantiles, gossips the inverse-variance-weighted releases for each quantile, combines the result with the anchor unless they conflict, multiplies by the default prior and averages its fits by its own evidence, exactly as above. Evaluating the score at the anchor makes it a control variate: a wrong anchor makes the release noisier, not biased. GHB-lite needs nothing beyond honest-but-curious

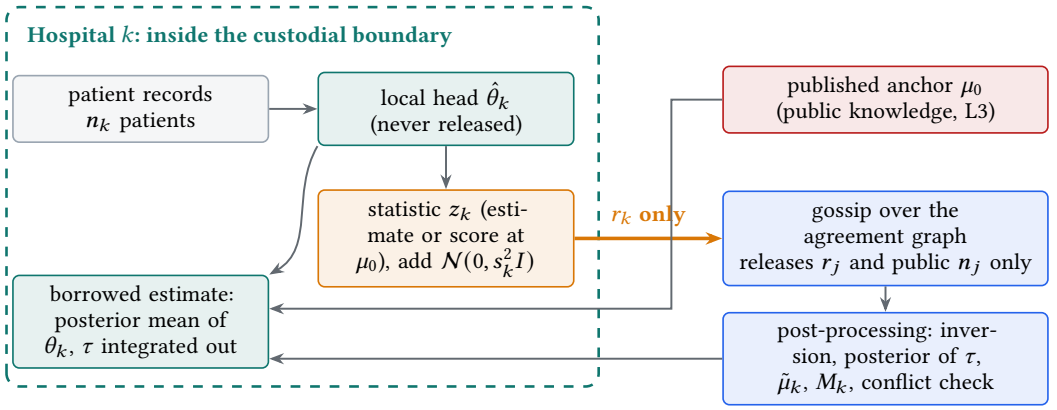



Fig. 4. The privacy boundary of GHB. Patient records and the local head stay inside the hospital. Only the hospital's masked share $r_k$ of the two network sums crosses the boundary; the masks cancel in the totals, which carry the privacy noise. Everything computed from the totals is post-processing (Proposition 5.1), and the hospital combines it with its own noise-free data locally.

neighbours, and it learns $\tau$ from the spread of the releases, which GHB cannot, since it releases no per-hospital quantity. Where the public feature distribution is exactly right and every hospital's release is informative, as in the simulations of Section 6, that makes GHB-lite the more accurate of the two. On real hospitals, neither condition held (Section 7).

### 5.4 The protocol and its privacy boundary

Algorithm 1 summarizes the protocol, and Figure 4 shows what crosses the custodial boundary: masked shares of two network sums, whose totals are privatized, and nothing else.

Proposition 5.1 (Privacy of GHB). *Suppose every patient's features satisfy $\|x\| \le G$, the secure summation reveals only the two totals, and the noise of each total has the stated variance. Then the pair $(\tilde{S}, \tilde{J})$ satisfies $\rho$-zero-concentrated differential privacy, and hence $(\epsilon, \delta)$-differential privacy with $\epsilon = \rho + 2\sqrt{\rho \ln(1/\delta)}$, with respect to replacing the record of any one patient at any hospital. Every model, estimate and message that GHB computes afterwards satisfies the same guarantee. If $c$ of the $K$ hospitals collude and remove their own noise shares from the totals, the guarantee for the other hospitals' patients weakens to $\rho K/(K-c)$.*

Proof. Replacing one record changes $\sum \psi_i$ by at most $2G$ and the upper triangle of $\sum \phi_i$ by at most $G^2/2$ in $\ell_2$ norm, as shown above. The Gaussian mechanism with noise $s$ and sensitivity $\Delta$ is $\Delta^2/(2s^2)$-zCDP [Bun and Steinke 2016], which is $\rho/2$ for each total; zCDP composes additively, so the pair is $\rho$-zCDP, and the conversion to $(\epsilon, \delta)$ is that of Bun and Steinke [2016]. Everything computed afterwards is a function of the two totals, of public quantities, and, at hospital $k$, of hospital $k$'s own data used only inside it and never released, so post-processing [Dwork and Roth 2014] gives the second claim; the networks' exchange of their estimates and the gossip of the masked shares are post-processing as well. Colluders who subtract their shares leave noise of variance $s^2(K-c)/K$, for which the same computation gives $\rho K/(K-c)$. □

The scope conditions of Section 4.2 carry over: a patient held by $m$ hospitals contributes $m$ records; site sizes are public; and every refit on the same patients is a new pair of releases, whose costs add under zero-concentrated accounting. A hospital's final model uses its own data without noise and is therefore not private. Like the local estimate in Proposition 4.1, it never leaves the

**Algorithm 1:** Graph Hierarchical Borrowing (one round of model fitting)

**Input:** graph $\mathcal{G}$ with gossip matrix $W$; privacy budget $(\epsilon, \delta)$, i.e. $\rho$; feature bound $G$; public encoder $f_{\phi_0}$; optionally a published model $\mu_0$ with stated uncertainty $\sigma_0^2$, and the estimates of other networks fitting the same head

**foreach** *hospital* $k$ *in parallel* **do**
- adapt $f_{\phi_0}$ to local unlabeled data with adapters (L1);
- compute $\sum_{i\in k} x_i(y_i - \frac{1}{2})$ and $\sum_{i\in k} x_i x_i^\top/4$; add pairwise masks agreed with neighbours and a share $\mathcal{N}(0, s^2/K)$ of the noise, $s_S = 2G/\sqrt{\rho}$, $s_J = G^2/(2\sqrt{\rho})$; `// the only released quantities`

gossip the masked shares until consensus: every hospital obtains $\tilde{S}$ and $\tilde{J}$ (Proposition 4.3); `// the masks cancel`

compute $\hat{w} = A\tilde{S}$ and $V_{\text{dp}} = s_S^2 AA^\top$, $A = (\tilde{J}^+ + I/s_0^2)^{-1}$; shrink $\hat{w}$ toward other networks' estimates, and combine it with $\mu_0$ unless they conflict; `// post-processing`

**foreach** *hospital* $k$ **do**
- compute locally $s_k$ at $\hat{w}$ and $J_k$; leave-one-out estimate $\mu_{-k} = \hat{w} - H_{-k}^{-1}s_k$, $V_{-k} = H_{-k}^{-1} + V_{\text{dp}}$;
- fit the local head under each prior $\mathcal{N}(\mu_{-k}, \text{diag}(\tau_q^2) + V_{-k}) \cdot \mathcal{N}(0, s^2 I)$, $q = 1, \ldots, Q$, $s \in \{0.5, 1, 2.5\}$, and under $\mathcal{N}(0, s^2 I)$ alone; average the fits weighted by the marginal likelihood of the local data (Remark 4.4); in the Gaussian case with known $\tau^2$ this is $\tilde{\theta}_k = \alpha_k^\star \hat{\theta}_k + (1 - \alpha_k^\star)\tilde{\mu}_k$ with $\alpha_k^\star$ of Proposition 4.1;
- calibrate with conformal prediction on local held-out data (L4); rank unlabeled patients by posterior variance (L5);

hospital; if it is published, its own privacy must be paid for separately. GHB-lite satisfies the same guarantee per hospital without any trust in summation, since its release is private on its own.

*Why this is a principled default.* The argument for GHB rests on results that are each well established, combined in a way that, to our knowledge, has not been set out for this setting. Separate estimation of three or more group means is inadmissible under squared error, and empirical Bayes shrinkage toward a common mean dominates it in the Gaussian model with equal variances [Efron and Morris 1973; Stein 1956]; Proposition 4.1 gives the best combination of a local and a population estimate in closed form when the population estimate is itself imperfect. Noise calibrated to a sum rather than to each contributor is the cheapest way to make that population estimate private [Bonawitz et al. 2017; Dwork and Roth 2014], and differential privacy is closed under post-processing, which makes everything after the two releases free. Gossip converges at a rate set by the spectral gap [Boyd et al. 2006; Xiao and Boyd 2004], which makes the summation possible without a server. Integrating the unknown heterogeneity out against each hospital's own evidence (Remark 4.4) lets borrowing switch itself off where it would hurt. And shared representations reduce the problem to a low-dimensional head [Maurer et al. 2016; Tripuraneni et al. 2020], which keeps the information matrix small enough to release. None of these is new. The case for GHB is that it is the simplest combination that meets all five requirements at once. Whether it also outperforms the alternatives has to be shown empirically, which Sections 6 and 7 do.

## 6 EXPERIMENTS

Sections 4 and 5 make claims under assumptions: the propositions take $\tau^2$ and $M$ as known, the model as Gaussian and every participant as honest. This section tests what is lost when those assumptions fail, in eighteen Monte Carlo simulations in which the truth is known, and Section 7

then asks the same questions of real hospitals, where it is not. Section 6.1 sets out the design. Section 6.2 checks that the theory survives estimation, over heterogeneity, hospital size, gossip rounds and failing links (Simulations 1 to 4 and 13). Section 6.3 leaves the Gaussian model, compares the method with federated, meta-analytic and decentralized baselines given every advantage, and measures what privacy, joint release and repeated releases cost (Simulations 5, 7, 8, 10, 14 and 15). Section 6.4 tries to break it with atypical hospitals, structured heterogeneity, a wrong anchor and dishonest hospitals (Simulations 6, 9, 11 and 12). Section 6.5 takes it apart, one component and one setting at a time (Simulations 16 and 17), and Section 6.6 measures what borrowing does to a protected group of patients (Simulation 18). Figure 5 puts every scenario on one axis, Table 2 lists the questions and the answers, and the online supplement gives the full design of each simulation.

### 6.1 Design of the simulation study

Unless stated otherwise, the network has 24 hospitals whose sizes are drawn once from a log-normal distribution with median 100 (16 to 479 patients), $\sigma^2 = 1$, $\delta = 10^{-5}$, $C = 1$ and $\epsilon = 1$, and the privatized releases are averaged by 60 rounds of gossip on the ring with two-hop links of Figure 2. *Plug-in GHB-lite* estimates everything it needs from the privatized releases and public quantities only: $\tau^2$ by DerSimonian-Laird [DerSimonian and Laird 1986] (Remark 4.4), the leave-one-out population estimate by inverse-variance weighting, and $M$ from those weights. Risk is reported relative to local estimation, $R_k/v_k$, so that values below 1 mean that a hospital gains. The code and every number are released with the paper.

*Which GHB is simulated.* The simulations were designed for, and run with, GHB-lite (Section 5.3), in which every hospital releases on its own: the Gaussian simulations validate the theory of Section 4, which is the theory of that protocol. GHB, with its joint releases (Section 5.2), was developed afterwards on real data, and was added to the logistic simulations that compare methods (8, 10, 11 and 15) without changing anything in them: the other methods' numbers are identical to those of the runs without it. In these simulated worlds GHB-lite has two advantages that real hospitals did not give it. The public feature distribution used to invert its releases is exactly the distribution of every hospital's features, and its releases, calibrated with 24 hospitals of about a hundred patients, are informative enough to estimate $\tau$. GHB cannot learn $\tau$ from per-hospital releases, because it makes none. The simulations measure what that costs where GHB-lite's assumptions hold exactly; Section 7 measures what happens where they do not.

*Evaluation protocol.* Three choices make the comparison hard for GHB-lite to pass. Every private method spends the same $(\epsilon, \delta)$ per hospital, and the iterative baselines are accounted with zero-concentrated DP, the tightest standard accounting. Every hyperparameter of a baseline (rounds, step size, fine-tuning steps, number of clusters, proximal weight) is chosen on separate replicates with access to the true loss, which no deployment has; GHB-lite has no tuning parameter. And GHB-lite itself was revised once, after a first run of Simulation 15 exposed weaknesses of an earlier version: the revision was developed on separate development seeds, the earlier version is reported alongside it, and four settings of Simulation 15 were never used during the revision. Results are means over Monte Carlo replicates with ±2 standard errors.

### 6.2 Does the theory survive estimation?

The propositions of Section 4 hold with $\tau^2$ and $M$ known. A real network must estimate both from privatized releases, so the first question is how much of the predicted gain survives that estimation, and for whom.

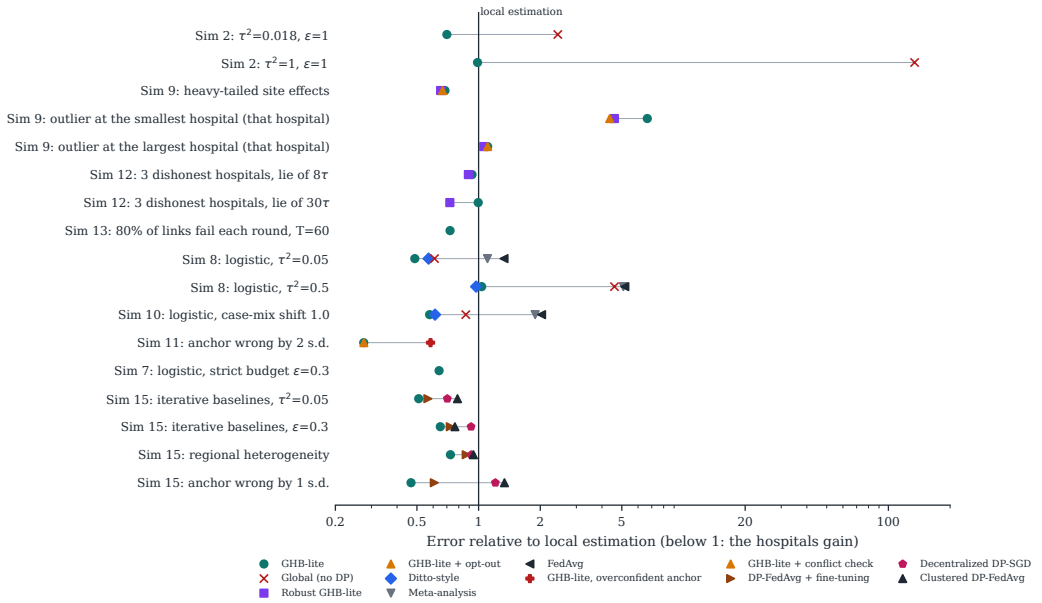



Fig. 5. Summary of Simulations 2 to 15. Each row is a scenario and each marker a method; the horizontal position is the method's error relative to local estimation (mean risk $R_k/v_k$ in the Gaussian simulations, mean excess log-loss divided by that of local estimation in the logistic ones). Left of the vertical line, the hospitals gain. Rows marked "that hospital" report the outlying hospital alone.

*Estimation costs little, and no hospital loses on average.* Simulation 1 confirms the closed forms behind Figure 1. Simulation 2 (Figure 6) varies the heterogeneity over four orders of magnitude. Plug-in GHB-lite stays at or below local estimation for every hospital, up to Monte Carlo error, while a single global model, which is the best choice when $\tau^2$ is tiny, becomes up to 134 times worse on average and 382 times worse for its worst hospital. The price of estimating $\tau^2$ is paid when heterogeneity is smallest: at $\tau^2 = 10^{-4}$ the plug-in reaches a mean relative risk of 0.12 against 0.07 with the true value. Integrating $\tau$ out (Remark 4.4) instead of plugging in an estimate keeps the worst hospital at 1.02 and costs slightly more at the smallest heterogeneity (0.14); its advantage appears with logistic heads and noisy releases, below. Gossip costs nothing once it has mixed. Simulation 4 shows that this takes 11 to 35 rounds on the topologies of Figure 2, far fewer than the worst-case bound of Proposition 4.3 requires, but that a hospital using the estimate after a single round on the ring is worse off than alone. Simulation 13 shows that mixing survives unreliable links: with 80% of links failing in every round, the risk after 60 rounds is 0.73, against 0.72 with exact averaging.

*The smallest hospitals gain most.* Simulation 3 (Figure 7, left) checks requirement (R4) hospital by hospital. Every hospital gains about the same number of virtual patients, $B \approx 46$, so those with 60 patients or fewer multiply their effective sample size by 2.28 on average and those with 300 or more by 1.08. A global model helps the small hospitals less (1.44) and cuts the effective sample size of the large ones to 0.15 of their own. With few hospitals, $\tau^2$ is poorly estimated (Figure 7, right). A weakly informative half-Cauchy prior on $\tau$, as Remark 4.4 recommends, helps from four hospitals upward, but with three hospitals most of the gain is lost.

Table 2. The eighteen simulations, the real-data confirmation, and their main results. The simulations run GHB-lite; where GHB was added, its value follows ("GHB"). Relative risk is $R_k/v_k$ and logistic results are mean excess log-loss (nats); below the local value, hospitals gain. $\epsilon = 1$ unless stated.

| Simulation | Question | Result |
|---|---|---|
| 1 Closed forms | Do Propositions 4.1 and 4.2 match simulation? | 98% of 101 comparisons within two standard errors; largest relative deviation 1.6% |
| 2 Heterogeneity | Is plug-in GHB-lite ever worse than local estimation? | Worst hospital at most 1.02 for $\tau^2 \in [10^{-4}, 1]$; a global model reaches 382 |
| 3 Equity, network size | Who gains, and how many hospitals are needed? | Gain in $n^{\text{eff}}$ 2.28 for small, 1.08 for large hospitals; with 3 hospitals relative risk 0.91 against 0.78 with known $\tau^2$ |
| 4 Gossip rounds | How many rounds are needed? | Within 2% of exact averaging after 11 to 35 rounds, against 73 to 203 from the worst-case bound |
| 5 Joint release | Does secure aggregation among small hospitals help? | Only if all hospitals are small: $B$ from 1.0 to 6.9 at $\epsilon = 0.1$; from 19.3 to 21.0 in a mixed network |
| 6 Regional structure | When does the GMRF prior of Equation (5) pay off? | Worse when regions barely differ (0.40 against 0.31), far better when they differ (0.40 against 0.94) |
| 7 Logistic heads | Does it work beyond the Gaussian model? | 0.0266 local, 0.0135 GHB-lite, 0.0173 pooled without privacy; 0.0175 against 0.0273 at $\epsilon = 0.3$ |
| 8 Federated baselines | Does it beat existing approaches at the same budget? | At $\tau^2 = 0.05$: GHB-lite 0.0134, GHB 0.0151, Ditto-style [Li et al. 2021a] 0.0156, meta-analysis 0.0303, FedAvg [McMahan et al. 2017] 0.0365; at $\tau^2 = 0.5$ GHB-lite 0.0282 against local 0.0273 |
| 9 Atypical hospitals | What happens to a hospital unlike the others? | Heavy tails: 0.69, robust prior 0.65; an outlier at the smallest hospital loses (6.7, at best 4.4), at the largest barely (1.11) |
| 10 Case-mix shift | Does it survive hospitals with different patients? | Shift of one s.d.: GHB-lite 0.0132, GHB 0.0142, Ditto-style [Li et al. 2021a] 0.0140, pooled 0.0198, local 0.0228 |
| 11 Wrong anchor | What if the published model is wrong? | Honest uncertainty or a conflict check: at most 0.0166 at any error; an over-confident anchor 0.0352 at 2 s.d.; GHB at most 0.0167 |
| 12 Dishonest hospitals | What if hospitals lie? | Plain GHB-lite never above 1.00; robust screening keeps 0.72 against large lies, at a cost of 0.81 against 0.73 when nobody lies |
| 13 Link failures | What if links fail during gossip? | With 80% of links failing each round, 0.73 after 60 rounds against 0.72 with exact averaging |
| 14 Repeated releases | What do refits on the same patients cost? | After 64 refits, $B = 1.5$ with basic composition and 32.4 with zero-concentrated accounting (one release: 46.1) |
| 15 Iterative federated learning | Does it beat multi-round training with the same budget and tuned hyperparameters? | At $\tau^2 = 0.05$: GHB-lite 0.0138, GHB 0.0159, DP-FedAvg [McMahan et al. 2018] with fine-tuning 0.0153, decentralized DP-SGD [Lian et al. 2017] 0.0191, clustered 0.0214; at $\epsilon = 0.3$ GHB-lite 0.0175 against 0.0196; worst hospital at most 1.06 for GHB-lite except at $\tau^2 = 0.5$, 1.39 to 2.96 for fine-tuning |
| 16 Ablation | What does each component contribute? | Error relative to GHB-lite without it: ridge release up to 1.39; $X^\top y$ release 1.74 under case-mix shift; plug-in $\tau^2$ up to 1.37 (posterior mean) and 1.33 (DerSimonian-Laird [DerSimonian and Laird 1986]); no conflict check 1.36 with a wrong anchor; equal weights up to 1.86; gossip and accounting free |
| 17 Sensitivity | How much do settings and conditions matter? | Settings: quantiles, public sample, threshold and rounds ($\geq$ 20) flat; prior scale of $\tau$ 0.90 to 1.17; misstated anchor up to 1.41. Conditions: error 0.40 to 0.69 of local, worst hospital at most 1.03 |
| 18 Group fairness | What does borrowing do to a protected group? | Minority excess log-loss ($\times 10^{-3}$) with a shared disparity: GHB-lite 17.8, local 34.3, FAIR-FATE [Salazar et al. 2023] 30.3; disparities that vary: GHB-lite 27.3, global 60.7; equalized odds of GHB-lite within 0.02 of the true models, FAIR-FATE closer to parity |
| Real data | Does it hold on real hospitals, against the literature tuned as a consortium could? | Pre-registered, five unseen TCGA networks: GHB 0.928 and 0.915 relative to local at $\epsilon = 1$ and 10; best literature method 0.976 and 0.919 (Section 7) |

### 6.3 Against federated and meta-analytic baselines

A method that gains over local estimation is useful only if it also gains over the alternatives a network could adopt instead. The comparisons below leave the Gaussian model for logistic-regression heads, and give every baseline the same privacy budget and hyperparameters tuned with access to the true loss.

*GHB-lite beats one-shot private baselines at moderate heterogeneity.* Simulations 7, 8 and 10 replace the Gaussian model by logistic-regression heads on four features. Each hospital releases its mean

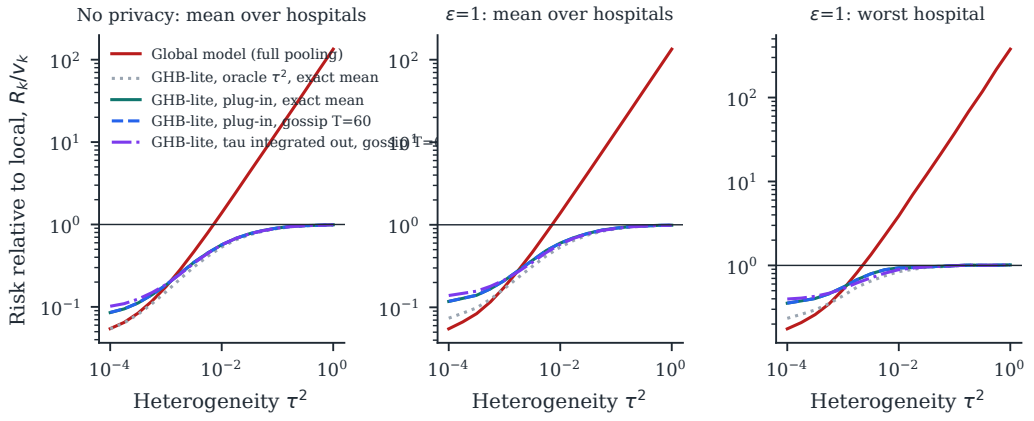



Fig. 6. Simulation 2: risk relative to local estimation against heterogeneity, for 24 hospitals of 16 to 479 patients. Plug-in GHB-lite estimates $\tau^2$ and $M$ from privatized releases averaged by gossip; the oracle uses the true $\tau^2$; the global model is the sample-size weighted mean of all site estimates, computed without privacy.

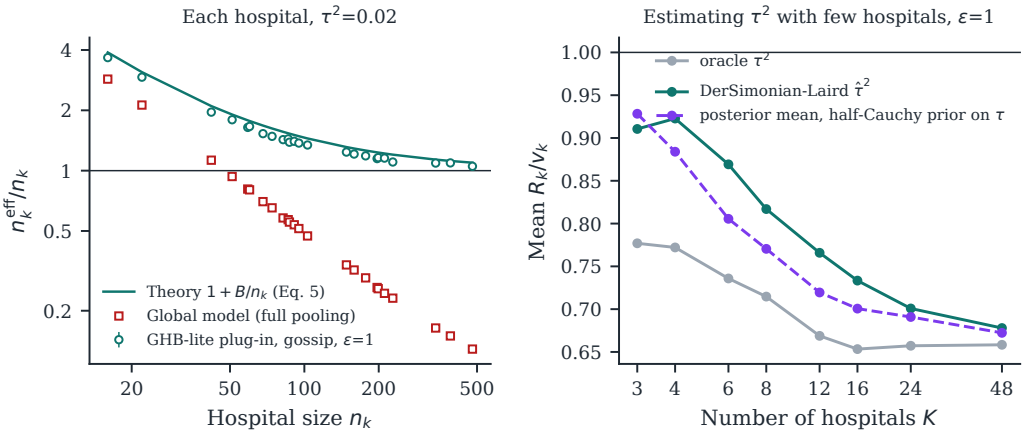



Fig. 7. Simulation 3. Left: effective sample size relative to own size for each hospital at $\tau^2 = 0.02$, $\epsilon = 1$; the line is $1 + B/n_k$ from Equation (3). Right: mean relative risk against the number of hospitals for three ways of obtaining $\tau^2$.

score at the published anchor $\mu_0$, with Gaussian noise calibrated to its sensitivity under zero-concentrated accounting (scope (d) of Section 4.2). The releases are mapped to the parameter scale by public moment matching, GHB-lite borrows coordinatewise, and each hospital integrates $\tau$ out (Remark 4.4), with the anchor entering its prior unless it conflicts with the network estimate (layer L3). Every private baseline consumes the same releases: one-shot federated averaging [McMahan et al. 2017], the random-effects meta-analytic mean, and a Ditto-style [Li et al. 2021a] personalized model that shrinks each local fit toward the federated average with a weight tuned on separate replicates, which favours it. At $\tau^2 = 0.05$ (Figure 8, left; Table 3), GHB-lite reaches an excess log-loss of 0.0134, 14% below the tuned Ditto-style [Li et al. 2021a] model (0.0156) and below a global model

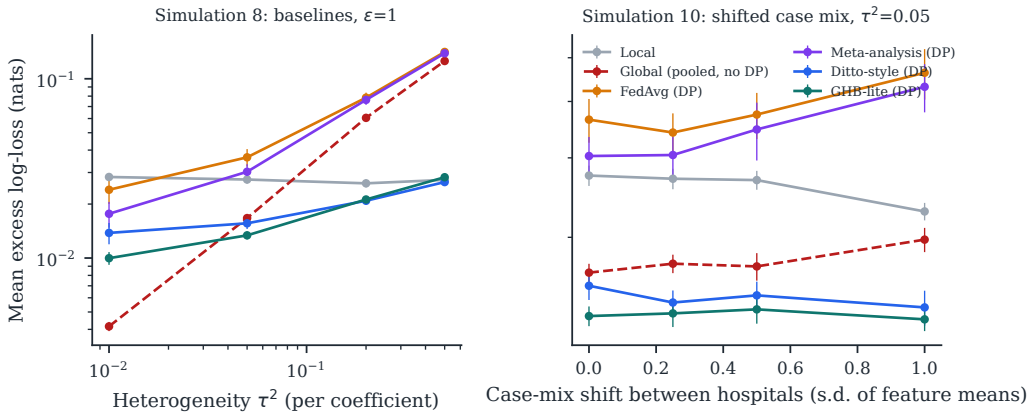



Fig. 8. Simulations 8 and 10, logistic-regression heads. Mean excess log-loss over the true model of GHB-lite, of private baselines that use the same releases (one-shot federated averaging [McMahan et al. 2017], random-effects meta-analysis, Ditto-style [Li et al. 2021a] personalization tuned on separate replicates) and of a global model trained on the pooled data without privacy. Left: against coefficient heterogeneity. Right: against case-mix shift between hospitals at $\tau^2 = 0.05$. Bars: ±2 standard errors.

Table 3. Simulations 8 and 10: exact values behind Figure 8. Values are mean excess log-loss in $10^{-3}$ nats, with two Monte Carlo standard errors in grey; the lowest private method in each row is bold. All private methods use the same releases at $\epsilon = 1$; the pooled model uses no privacy.

| Setting | Local | GHB | GHB-lite | Ditto-style [Li et al. 2021a] | Meta-analysis [DerSimonian and Laird 1986] | FedAvg [McMahan et al. 2017] | Pooled, no DP |
|---|---|---|---|---|---|---|---|
| $\tau^2 = 0.01$ | 28.3 (1.1) | 11.7 (0.9) | **10.0** (0.8) | 13.8 (1.9) | 17.7 (2.9) | 24.0 (4.0) | 4.2 (0.2) |
| $\tau^2 = 0.05$ | 27.4 (1.4) | 15.1 (0.9) | **13.4** (0.7) | 15.6 (1.1) | 30.3 (3.1) | 36.5 (4.1) | 16.7 (0.8) |
| $\tau^2 = 0.2$ | 26.1 (1.3) | 22.9 (1.1) | 21.2 (0.9) | **20.9** (0.9) | 76.3 (4.7) | 78.4 (5.2) | 60.5 (2.3) |
| $\tau^2 = 0.5$ | 27.3 (1.4) | 30.7 (1.1) | 28.2 (1.3) | **26.5** (1.1) | 138.8 (5.1) | 141.1 (5.2) | 125.6 (4.3) |
| shift 0.25 | 27.0 (1.4) | 15.3 (0.8) | **13.6** (0.9) | 14.3 (0.9) | 30.4 (3.0) | 34.1 (3.5) | 17.5 (0.8) |
| shift 0.5 | 26.8 (1.3) | 15.3 (0.8) | **13.8** (1.0) | 14.9 (1.1) | 34.7 (5.1) | 37.4 (4.4) | 17.2 (1.2) |
| shift 1.0 | 22.8 (1.0) | 14.2 (1.0) | **13.2** (0.8) | 14.0 (1.2) | 43.1 (5.3) | 46.3 (5.9) | 19.8 (1.2) |

trained on the pooled data without any privacy (0.0167); federated averaging [McMahan et al. 2017] and meta-analysis, which give every hospital the same model, are worse than local estimation (0.0365 and 0.0303 against 0.0274). When hospitals also differ in case mix (Simulation 10, right), GHB-lite keeps its advantage (0.0132 against 0.0140 for Ditto-style [Li et al. 2021a] and 0.0198 for the pooled model). From $\tau^2 = 0.2$ on, the tuned Ditto-style model is slightly more accurate, and at $\tau^2 = 0.5$ it is the only method that gains at all: GHB-lite reaches 0.0282, local estimation 0.0273 and the Ditto-style model 0.0265. GHB, which makes no per-hospital release and so cannot learn $\tau$ from the spread of the releases, is less accurate than GHB-lite in this world (0.0151 at $\tau^2 = 0.05$, 0.0142 under case-mix shift). It is still more accurate than the tuned Ditto-style [Li et al. 2021a] model at $\tau^2 = 0.05$ and than every model that gives all hospitals the same parameters, and at $\tau^2 = 0.5$ it is worse than local estimation (0.0307), as GHB-lite is.

*Against iterative federated learning.* One-shot baselines are weak by construction, so Simulation 15 compares GHB-lite with multi-round training, taking the strongest competitor in each family closest to it: DP-FedAvg [McMahan et al. 2017, 2018] with and without local fine-tuning; decentralized

DP-SGD [Lian et al. 2017] on the same agreement graph, which, like GHB-lite, needs no server [Lian et al. 2017; Warnat-Herresthal et al. 2021]; and clustered federated learning in the style of IFCA [Ghosh et al. 2020], made private by assigning clusters from one privatized release. Every method spends the same $(\epsilon, \delta)$ per hospital, accounted with zero-concentrated DP, and the baselines' rounds, step size, fine-tuning steps and number of clusters are chosen on 20 separate replicates with access to the true loss. GHB-lite has no tuning parameter. A first run of this simulation exposed the weaknesses of an earlier version of GHB-lite, which released ridge fits and plugged in an estimate of $\tau^2$; we revised it on separate development seeds into the version of Section 5.3. Figure 9 and Tables 4 and 5 report both, and four of the ten settings (marked $*$) were never used during the revision.

GHB-lite is more accurate than every tuned baseline in six of the ten settings, including three of the four held out: with nearly identical hospitals (0.0101 against 0.0114 for DP-FedAvg [McMahan et al. 2018] with fine-tuning, the strongest federated competitor), at $\tau^2 = 0.05$ (0.0138 against 0.0153), at a strict budget (0.0175 against 0.0196 at $\epsilon = 0.3$), with regional heterogeneity (0.0209 against 0.0212 for Ditto-style [Li et al. 2021a]), at $\tau^2 = 0.1$ and $\epsilon = 0.5$ (0.0184 against 0.0217 for DP-FedAvg with fine-tuning), and with 48 hospitals (0.0144 against 0.0146). In three settings the Ditto-style [Li et al. 2021a] model, which shrinks each hospital toward the federated average of GHB-lite's own releases with a weight tuned on the true loss, is slightly more accurate: at $\tau^2 = 0.2$ (0.0205 against 0.0217), without privacy (0.0101 against 0.0104) and with a wrong anchor (0.0148 against 0.0162). At $\tau^2 = 0.5$ no private method beats local estimation (0.0260; GHB-lite 0.0275). GHB-lite is more accurate than DP-FedAvg [McMahan et al. 2018] with fine-tuning, decentralized DP-SGD [Lian et al. 2017] and clustered training in every private setting.

The gain at a strict budget comes from borrowing, not only from the anchor. A control that uses no other hospital, a few gradient steps on the hospital's own data starting from the published anchor, with the number of steps tuned, reaches 0.0205 at $\epsilon = 0.3$, against 0.0175 for GHB-lite. When the anchor is wrong by one standard deviation per coefficient, that control becomes worse than local estimation (0.0419 against 0.0346), and so do decentralized and clustered training, which start from it (0.0418 and 0.0463), while GHB-lite, which drops the conflicting anchor, keeps 0.0162.

The methods differ most in who pays. GHB-lite's worst hospital stays within a factor 1.06 of its local error in every private setting except $\tau^2 = 0.5$ (1.15), as Proposition 4.1 and Remark 4.4 intend. Fine-tuning leaves some hospital worse off in every private setting, by a factor of 1.39 to 2.96, and those who pay are the large hospitals, whose own data are overridden by the federated model: at $\epsilon = 0.3$ the eight largest go from 0.0094 alone to 0.0131 with fine-tuning, and to 0.0079 with GHB-lite. What remains is the cost of estimating heterogeneity from noisy releases: with the true $\tau^2$, GHB-lite would reach 0.0158 at $\epsilon = 0.3$ and 0.0081 at $\tau^2 = 0.01$. The revision closed most of that gap; the earlier version reached 0.0259 at $\epsilon = 0.3$, barely below local estimation. In short, at the same budget GHB-lite is the most accurate private method or within 10% of the most accurate in every setting where any method gains, and it is the only one whose worst hospital stays close to its local error.

GHB, added to this simulation without changing anything else, is more accurate than every iterative baseline (DP-FedAvg [McMahan et al. 2018] with fine-tuning, decentralized DP-SGD [Lian et al. 2017] and clustered training) in 4 of the ten settings, and more accurate than GHB-lite in one, the strictest budget ($\epsilon = 0.3$: 0.0168 against 0.0175); at $\tau^2 = 0.05$ it reaches 0.0159 against 0.0138. Its worst hospital stays within a factor 1.15 of its local error in every private setting except $\tau^2 = 0.5$. GHB is more accurate than GHB-lite only at the strictest budget, where per-hospital noise is largest, and without privacy it is far less accurate (0.0161 against 0.0104), because it cannot learn $\tau$ from releases. In the other private settings it is within 11% of the tuned DP-FedAvg [McMahan et al. 2018] with fine-tuning, and the Ditto-style [Li et al. 2021a] model, which shrinks toward the

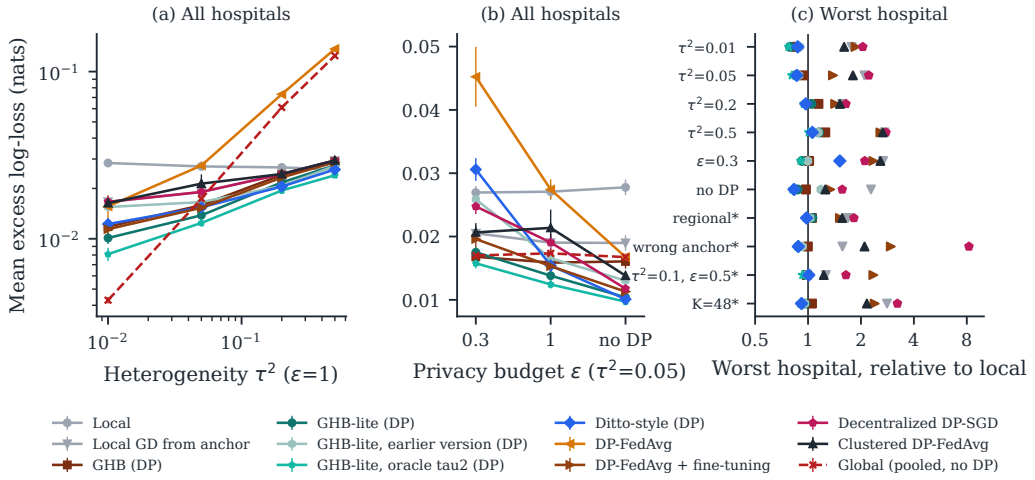



Fig. 9. Simulation 15: GHB-lite against iterative federated learning at the same privacy budget (logistic heads, 24 hospitals, mean excess log-loss over 60 replicates, ±2 standard errors). Baseline hyperparameters are tuned with access to the true loss; GHB-lite is not tuned. (a) Against heterogeneity at $\epsilon = 1$. (b) Against the privacy budget at $\tau^2 = 0.05$. (c) The worst hospital's error relative to its own local estimate in each setting; right of 1, some hospital is worse off than alone; settings marked $*$ were not used to revise GHB-lite. "GHB-lite, earlier version" is the version before the revision; "GHB-lite, oracle $\tau^2$" uses the true heterogeneity and is a reference, not a method.

Table 4. Simulation 15: exact values behind Figure 9. Values are mean excess log-loss in $10^{-3}$ nats, with two Monte Carlo standard errors in grey; the lowest private method in each row is bold. Settings marked $*$ were not used while revising GHB-lite. FT: fine-tuning.

| Setting | Local | Anchor GD | GHB | GHB-lite | Ditto-style [Li et al. 2021a] | DP-FedAvg + FT [McMahan et al. 2018] | Decentr. DP-SGD [Lian et al. 2017] | Clustered DP-FedAvg [Ghosh et al. 2020] | GHB-lite, true $\tau^2$ | Pooled, no DP |
|---|---|---|---|---|---|---|---|---|---|---|
| $\tau^2 = 0.01$ | 28.4 (1.3) | 16.6 (1.5) | 11.8 (0.7) | **10.1** (0.8) | 12.2 (0.8) | 11.4 (1.5) | 16.6 (1.4) | 16.3 (1.3) | 8.1 (0.7) | 4.3 (0.2) |
| $\tau^2 = 0.05$ | 27.1 (1.2) | 19.0 (1.4) | 15.9 (0.9) | **13.8** (0.8) | 15.4 (1.1) | 15.3 (0.8) | 19.1 (1.4) | 21.4 (2.9) | 12.4 (0.6) | 17.3 (0.7) |
| $\tau^2 = 0.2$ | 26.7 (1.3) | 24.4 (1.1) | 24.1 (1.1) | 21.7 (0.9) | **20.5** (0.8) | 23.2 (1.1) | 24.4 (1.1) | 24.4 (1.1) | 19.4 (0.8) | 61.0 (1.9) |
| $\tau^2 = 0.5$ | **26.0** (1.1) | 29.3 (1.2) | 28.8 (1.1) | 27.5 (1.1) | 26.0 (1.1) | 28.7 (1.2) | 29.4 (1.2) | 29.4 (1.5) | 24.0 (0.9) | 124.4 (3.2) |
| $\epsilon = 0.3$ | 26.9 (1.0) | 20.5 (1.4) | **16.8** (1.1) | 17.5 (0.8) | 30.6 (1.8) | 19.6 (1.3) | 24.8 (1.2) | 20.6 (1.5) | 15.8 (0.8) | 17.0 (0.6) |
| no DP | 27.8 (1.2) | 19.0 (1.2) | 16.1 (1.1) | 10.4 (0.3) | **10.1** (0.3) | 11.3 (0.5) | 11.8 (0.5) | 13.8 (0.6) | 9.7 (0.3) | 16.7 (0.5) |
| regional* | 28.7 (1.2) | 26.5 (1.2) | 23.4 (0.9) | **20.9** (0.9) | 21.2 (0.8) | 24.9 (1.1) | 26.4 (1.2) | 27.1 (1.6) | 19.9 (0.7) | 48.5 (3.1) |
| wrong anchor* | 34.6 (2.2) | 41.9 (2.7) | 16.7 (1.0) | 16.2 (1.2) | **14.8** (1.1) | 21.0 (2.1) | 41.8 (4.5) | 46.3 (4.3) | 15.0 (1.2) | 16.9 (0.6) |
| $\tau^2 = 0.1$, $\epsilon = 0.5$* | 26.9 (1.2) | 22.8 (1.1) | 19.6 (0.9) | **18.4** (0.8) | 23.8 (1.5) | 21.7 (1.3) | 23.9 (1.2) | 22.9 (1.1) | 17.1 (0.7) | 32.7 (1.2) |
| $K = 48$* | 32.4 (1.0) | 22.4 (1.6) | 17.7 (0.6) | **14.4** (0.5) | 14.6 (0.6) | 16.0 (0.8) | 22.5 (1.3) | 24.1 (2.9) | 13.2 (0.5) | 17.2 (0.4) |

average of GHB-lite's releases with a weight tuned on the true loss, is more accurate than GHB in seven of the ten settings. In these worlds, where GHB-lite's inversion is exact, the per-hospital releases are worth more than their noise; on real hospitals they were not (Section 7).

*Privacy is the binding constraint.* In Simulation 7, GHB-lite reduces the excess log-loss of local estimation by 49% at $\epsilon = 1$, reaches 0.0106 without privacy, and keeps most of its gain at $\epsilon = 0.3$ (0.0175 against 0.0273). The earlier version, which released ridge fits, kept almost none at that budget (0.0259 against 0.0269 in Simulation 15): what a hospital releases matters as much as how much privacy it spends. The privacy cost that the Gaussian analysis locates in $M$ remains real, and for small hospitals it is large. Joint release through secure aggregation (Simulation 5) recovers

Table 5. Simulation 15: who pays. Left: the largest ratio of any hospital's mean excess log-loss to its own local value (above 1, some hospital is worse off than alone; lowest in bold). Middle: the percentage of hospitals that gain over local estimation. Right: the mean excess log-loss of the eight largest hospitals ($10^{-3}$ nats).

| | Worst hospital, relative to local | | | | | | | | Hospitals that gain (%) | | | Eight largest | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Setting | Local | Anchor GD | GHB | GHB-lite | Ditto [Li et al. 2021a] | FedAvg + FT [McMahan et al. 2018] | Decentr. [Lian et al. 2017] | Clust. [Ghosh et al. 2020] | GHB-lite | Ditto | FedAvg + FT | Local | GHB-lite | FedAvg + FT |
| $\tau^2 = 0.01$ | 1.00 | 1.77 | 0.85 | **0.79** | 0.87 | 1.87 | 2.05 | 1.60 | 100 | 100 | 88 | 10.3 | 6.6 | 9.9 |
| $\tau^2 = 0.05$ | 1.00 | 2.10 | 0.94 | **0.85** | 0.86 | 1.39 | 2.22 | 1.80 | 100 | 100 | 88 | 10.0 | 7.9 | 9.8 |
| $\tau^2 = 0.2$ | 1.00 | 1.56 | 1.15 | 1.04 | **0.97** | 1.43 | 1.64 | 1.52 | 83 | 100 | 71 | 10.1 | 9.8 | 11.1 |
| $\tau^2 = 0.5$ | **1.00** | 2.72 | 1.25 | 1.15 | 1.06 | 2.58 | 2.77 | 2.66 | 12 | 46 | 25 | 9.5 | 9.7 | 15.6 |
| $\epsilon = 0.3$ | 1.00 | 2.67 | 1.01 | **0.92** | 1.52 | 2.35 | 2.10 | 2.58 | 100 | 0 | 62 | 9.4 | 7.9 | 13.1 |
| no DP | 1.00 | 2.28 | 0.97 | 0.87 | **0.83** | 1.34 | 1.57 | 1.25 | 100 | 100 | 92 | 10.6 | 6.9 | 7.9 |
| regional* | 1.00 | 1.69 | 1.05 | 1.06 | **0.98** | 1.52 | 1.82 | 1.56 | 92 | 100 | 71 | 10.0 | 9.3 | 11.2 |
| wrong anchor* | 1.00 | 1.57 | 1.00 | 0.88 | **0.88** | 2.96 | 8.23 | 2.10 | 100 | 100 | 58 | 9.6 | 8.0 | 14.6 |
| $\tau^2 = 0.1$, $\epsilon = 0.5$* | 1.00 | 1.26 | 0.98 | **0.97** | 1.01 | 2.35 | 1.64 | 1.23 | 100 | 92 | 58 | 10.2 | 9.1 | 16.2 |
| $K = 48$* | 1.00 | 2.81 | 1.05 | 0.96 | **0.92** | 2.39 | 3.22 | 2.16 | 100 | 100 | 77 | 11.2 | 8.2 | 12.2 |

part of it only in networks where every hospital is small; where some hospitals are large, they dominate the population estimate anyway, and with logistic heads groups of three gained at most 12% in development runs (supplement). Repeated releases compound the cost (Simulation 14): a hospital that refits its head 64 times on the same patients keeps $B = 32.4$ borrowed patients under zero-concentrated accounting [Bun and Steinke 2016], against 1.5 under basic composition and 46.1 for a single release, so refits should be accounted with zCDP or made on new patients.

### 6.4 What breaks it

The comparisons so far assume a well-behaved network. The next simulations remove that assumption on purpose, one failure at a time.

*Atypical hospitals, wrong anchors and lies.* Simulations 9, 11 and 12 are adversarial. *Atypical hospitals* (Simulation 9) are the failure that Proposition 4.1 predicts. With heavy-tailed site effects the mean risk stays low (0.69), and a Student-$t$ prior, which changes only the hospital's own computation, lowers it to 0.65 at a small cost when effects are Gaussian (0.76 against 0.72). A single outlying hospital placed six standard deviations away is a different matter. If it is the largest hospital, its own estimate is precise, the network's $\hat{\tau}^2$ grows, and it barely loses (1.11). If it is the smallest, it loses badly (6.7 times its local risk), and neither the robust prior nor a rule that keeps the local estimate when it is three standard errors from the network estimate brings it below 4.4. A small hospital cannot tell from its own data that it is atypical; only external knowledge, such as the structured prior of Section 4.5 (Simulation 6), can. *A wrong anchor* (Simulation 11) is harmless when its uncertainty is stated honestly or when the anchor is dropped on conflict with the network estimate (at most 0.0166 at any error), but an overconfident anchor two standard deviations off raises the excess log-loss to 0.0352. GHB, which combines the anchor with its network estimate only when the two agree, stays at or below 0.0167 at every error. This simulation also led to two design rules. A wrong anchor must not bias the releases: the score is evaluated at $\mu_0$, but its inversion is exact moment matching, and its linearization points are free of the anchor or have passed the conflict check. And the conflict check, which costs nothing when the anchor is right, is part of both protocols rather than an option. *Dishonest hospitals* (Simulation 12) cannot push honest hospitals below local performance in any scenario we tried: a large lie inflates $\hat{\tau}^2$, which switches borrowing off (worst relative risk 1.00). Screening the releases for implausible values before pooling, which is post-processing and costs no privacy, keeps the gain against large lies (0.72) at a cost when nobody lies (0.81 against 0.73). Opt-out and screening remain options of GHB-lite.

### 6.5 Which components and settings matter

A method with many parts invites two questions: whether every part earns its place, and whether the results depend on settings chosen with knowledge of them. Simulations 16 and 17 answer both.

*What each component contributes.* Simulation 16 removes or replaces one component of GHB-lite at a time in eight settings of the logistic world, with every variant computed from the same data and the same noise draw, so that differences are paired (Figure 10 and Table 6; 80 replicates per setting). Every component earns its place somewhere, and none costs much where it is not needed. The score release matters against both alternatives, in different ways. A ridge fit is significantly worse in seven of the eight settings, by up to a factor 1.39. Releasing the sufficient statistic $X^\top y$ instead of the score is equivalent except under case-mix shift, where it is 1.74 times worse: evaluating the score at the anchor removes most of the dependence of the release on the hospital's own feature distribution, which the public moments cannot know. Inverse-variance weighting of the network mean is worth up to a factor 1.86, most when hospitals are similar or the anchor is wrong. The anchor lowers the error by up to a factor 1.90 at a strict budget, and the conflict check is free when the anchor is right and worth a factor 1.36 when it is wrong.

The rule for the heterogeneity is where the design is most delicate. A plug-in posterior median of $\tau^2$, or integrating $\tau$ out without weighting by each hospital's own evidence, is slightly better when hospitals are nearly identical (down to 0.98 and 0.97), but 1.19 and 1.17 times worse when they differ strongly. The plug-in posterior mean is up to 1.37 times worse at a strict budget, and DerSimonian-Laird [DerSimonian and Laird 1986] up to 1.33 times worse. Integrating $\tau$ out with each hospital's evidence is within 4% of the best of these rules in every setting, and no other rule is: that is why GHB-lite uses it. Re-linearizing the releases (at most 1.03) and the default prior (at most 1.02) are small refinements. Two components cost nothing measurable. Gossip with 60 rounds matches exact averaging by a server, and for a single release zero-concentrated accounting matches the classical Gaussian mechanism to within one percent; its value lies in repeated releases (Simulation 14). What remains is the heterogeneity itself: knowing $\tau^2$ would lower the error by a further factor between 0.79 and 0.96. The same ablation in the Gaussian model, in closed form, shows the same pattern (supplement): at $\epsilon = 0.3$ and $\tau^2 = 0.02$, DerSimonian-Laird [DerSimonian and Laird 1986] reaches a relative risk of 0.91 against 0.74 for the integrated rule, and when hospitals are nearly identical equal weights reach 0.66 against 0.36.

*How sensitive GHB-lite is to its settings.* Simulation 17 varies each setting of GHB-lite over a wide range, the others at their reported values, in three base settings (Figure 11, Table 7). Most settings hardly matter. The number of posterior quantiles of $\tau$ is flat from three upward, and a single quantile, the plug-in median, costs up to a factor 1.06. The size of the public sample used for the moments of the inversion has no measurable effect from 2,000 draws. The conflict threshold is flat from 2.5 upward. Gossip needs about 20 rounds on this graph, after which the error is within 1.5% of its limit; with 5 rounds it is up to 1.18 times higher. Two settings matter. The scale of the half-Cauchy prior on $\tau$ trades the two ends of the heterogeneity range against each other: a small scale helps nearly identical hospitals at a strict budget (down to 0.90) and hurts heterogeneous ones, and a large scale does the reverse (up to 1.17); the reported 0.5 is within 11% of the best scale in every base setting, while either extreme loses up to 17%. The stated uncertainty of the anchor matters most: overstating its accuracy (0.1 when the true error is 0.3) costs up to 21%, and understating it costs up to a factor 1.41 at a strict budget, where the anchor carries most of the information. As Simulation 11 found, the anchor's uncertainty should be stated honestly. A tighter default prior on the coefficients helps in this world (down to 0.90 with a standard deviation of 1), because its true coefficients are smaller than 1 in magnitude; we keep the generic default of 2.5

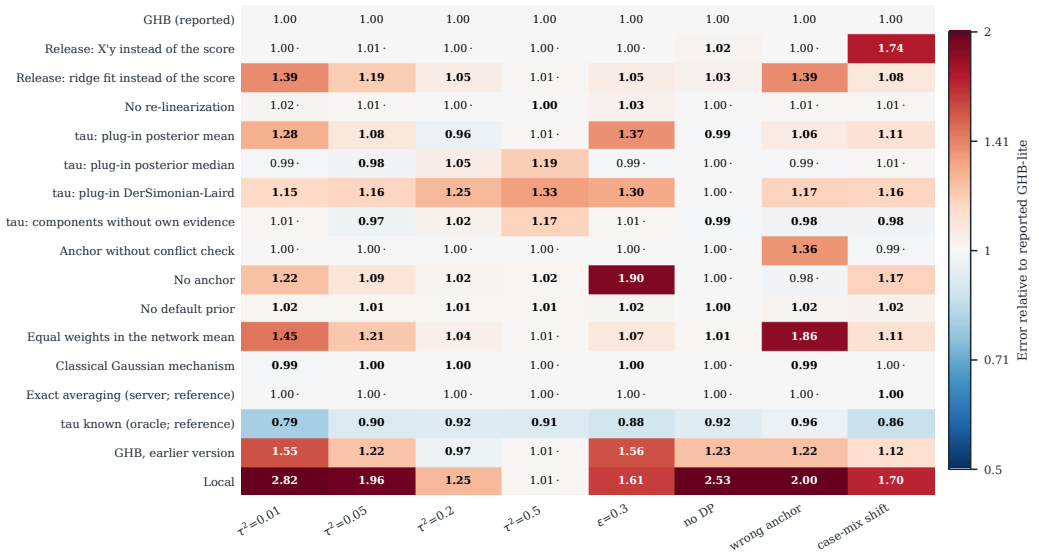



Fig. 10. Simulation 16: ablation of GHB-lite (logistic heads, 24 hospitals, 80 replicates per setting). Each cell is the mean excess log-loss of a variant divided by that of GHB-lite as reported, computed on the same data and noise; above 1 (red), removing or replacing the component hurts. Bold: paired difference larger than two standard errors; a trailing dot marks differences within Monte Carlo error. The last three rows are references.

Table 6. Simulation 16: exact values behind Figure 10. First row: mean excess log-loss of GHB-lite as reported ($10^{-3}$ nats). Other rows: the mean excess log-loss of each variant divided by that of GHB-lite on the same data and noise (80 replicates per setting); bold where the paired difference exceeds two standard errors. The last rows are references.

| Variant | $\tau^2 = 0.01$ | $\tau^2 = 0.05$ | $\tau^2 = 0.2$ | $\tau^2 = 0.5$ | $\epsilon = 0.3$ | no DP | wrong anchor | shift |
|---|---|---|---|---|---|---|---|---|
| Reported GHB-lite ($10^{-3}$ nats) | 9.8 | 14.0 | 21.6 | 27.6 | 17.1 | 11.1 | 16.1 | 13.5 |
| Release: $X^\top y$ instead of the score | 1.00 | 1.01 | 1.00 | 1.00 | 1.00 | **1.02** | 1.00 | **1.74** |
| Release: ridge fit instead of the score | **1.39** | **1.19** | **1.05** | 1.01 | **1.05** | **1.03** | **1.39** | **1.08** |
| No re-linearization | 1.02 | 1.01 | 1.00 | **1.00** | **1.03** | 1.00 | 1.01 | 1.01 |
| $\tau$: plug-in posterior mean | **1.28** | **1.08** | **0.96** | 1.01 | **1.37** | **0.99** | **1.06** | **1.11** |
| $\tau$: plug-in posterior median | 0.99 | **0.98** | **1.05** | **1.19** | 0.99 | 1.00 | 0.99 | 1.01 |
| $\tau$: plug-in DerSimonian-Laird | **1.15** | **1.16** | **1.25** | **1.33** | **1.30** | 1.00 | **1.17** | **1.16** |
| $\tau$: components without own evidence | 1.01 | **0.97** | **1.02** | **1.17** | 1.01 | **0.99** | **0.98** | **0.98** |
| Anchor without conflict check | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | **1.36** | 0.99 |
| No anchor | **1.22** | **1.09** | **1.02** | **1.02** | **1.90** | 1.00 | 0.98 | **1.17** |
| No default prior | **1.02** | **1.01** | **1.01** | **1.01** | **1.02** | **1.00** | **1.02** | **1.02** |
| Equal weights in the network mean | **1.45** | **1.21** | **1.04** | 1.01 | **1.07** | **1.01** | **1.86** | **1.11** |
| Classical Gaussian mechanism | **0.99** | **1.00** | **1.00** | 1.00 | **1.00** | 1.00 | **0.99** | 1.00 |
| Exact averaging (server; reference) | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | **1.00** |
| $\tau$ known (oracle; reference) | **0.79** | **0.90** | **0.92** | **0.91** | **0.88** | **0.92** | **0.96** | **0.86** |
| GHB-lite, earlier version | **1.55** | **1.22** | **0.97** | 1.01 | **1.56** | **1.23** | **1.22** | **1.12** |
| Local | **2.82** | **1.96** | **1.25** | 1.01 | **1.61** | **2.53** | **2.00** | **1.70** |

[Gelman et al. 2008] rather than tune it to the simulation, and dropping the default prior costs at most 1.03.

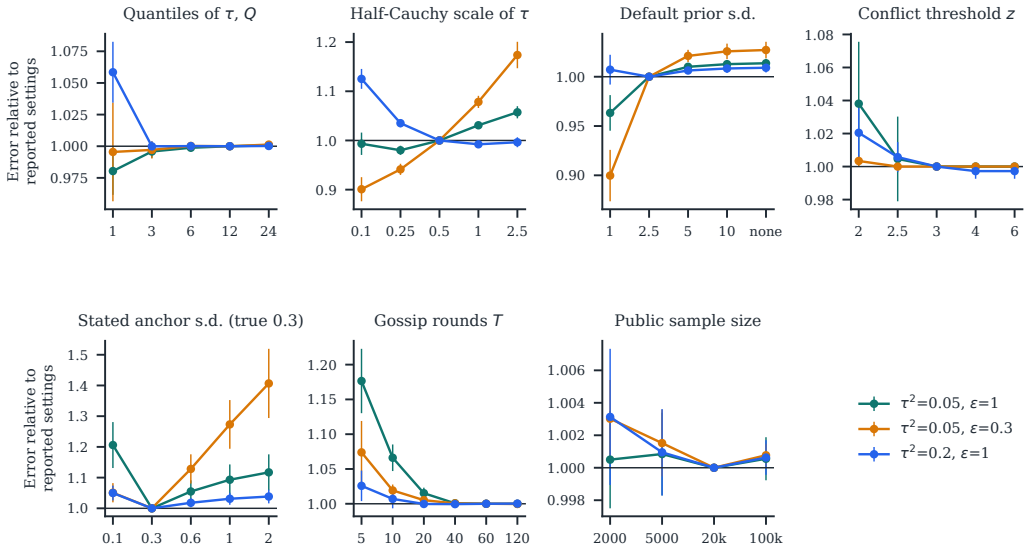



Fig. 11. Simulation 17, part A: sensitivity of GHB-lite to its own settings (logistic heads, 40 paired replicates per point). Each point is the mean excess log-loss with the setting changed, divided by that with the reported settings ($Q = 12$, scale 0.5, default prior s.d. 2.5, $z = 3$, stated anchor s.d. 0.3, $T = 60$, 20,000 public draws). Bars: ±2 standard errors of the paired ratio.

*How sensitive GHB-lite is to the network.* The second part of Simulation 17 varies the conditions that GHB-lite does not control (Figure 12, Table 8). For 6 to 96 hospitals, median sizes from 25 to 400 patients, budgets from $\epsilon = 0.1$ to no privacy, $\delta$ from $10^{-3}$ to $10^{-9}$ and all four topologies of Figure 2, GHB-lite's error stays between 0.40 and 0.69 times that of local estimation, and its worst hospital within a factor 1.03 of its own local error. The gain grows with the number of hospitals up to about 48 (0.59 with 6, 0.43 with 48) and with the budget (0.69 at $\epsilon = 0.1$, 0.40 without privacy). It is largest for networks of hospitals of about 100 patients, and smaller both for very small hospitals, whose releases are the noisiest, and for large ones, which need less (0.55 at a median of 25, 0.66 at 400). Neither $\delta$ (0.48 to 0.55) nor the topology (0.49 to 0.53) matters much once gossip has mixed. Differences of about 0.03 between repeats of the same base setting are Monte Carlo variation, since each condition uses its own seeds.

## 6.6 Fairness to groups of patients

Every simulation so far measures fairness across hospitals: whether some hospital pays for the others' gain. The last one turns to fairness across patients.

*What borrowing does to a protected group.* Simulation 18 asks the question of Section 2.3 that the other simulations cannot: how GHB-lite treats a group of patients defined by a protected attribute. One of the four features becomes a binary attribute $S$ (the minority group, $S = 1$), whose share varies across hospitals (Beta-distributed with mean 0.3, or 0.1 when the minority is rare), so that the group is scarce at some hospitals, as in the heterogeneous splits of FAIR-FATE's experiments [Salazar et al. 2023]. The coefficient of $S$ is each hospital's true disparity in outcome risk, on the log-odds scale. Four scenarios vary how much hospitals agree about it: a shared disparity ($\mu_S = 0.5$, $\tau_S^2 = 0.05$, like every other coefficient), a disparity that varies ten times more than the other coefficients

Table 7. Simulation 17, part A: exact values behind Figure 11. Mean excess log-loss with one setting changed, divided by that with the reported settings (italic), on the same data and releases; two standard errors of the paired ratio in grey (40 replicates).

| Setting | Value | $\tau^2 = 0.05, \epsilon = 1$ | $\tau^2 = 0.05, \epsilon = 0.3$ | $\tau^2 = 0.2, \epsilon = 1$ |
|---|---|---|---|---|
| Quantiles of $\tau$, $Q$ | 1 | 0.980 (0.019) | 0.995 (0.039) | 1.058 (0.024) |
| | 3 | 0.996 (0.005) | 0.997 (0.007) | 1.000 (0.004) |
| | 6 | 0.999 (0.002) | 1.000 (0.002) | 1.000 (0.002) |
| | *12* | 1 | 1 | 1 |
| | 24 | 1.001 (0.001) | 1.001 (0.002) | 1.000 (0.001) |
| Half-Cauchy scale of $\tau$ | 0.1 | 0.993 (0.023) | 0.901 (0.025) | 1.125 (0.020) |
| | 0.25 | 0.980 (0.009) | 0.941 (0.011) | 1.035 (0.008) |
| | *0.5* | 1 | 1 | 1 |
| | 1.0 | 1.031 (0.007) | 1.078 (0.012) | 0.992 (0.006) |
| | 2.5 | 1.057 (0.012) | 1.174 (0.027) | 0.996 (0.010) |
| Default prior s.d. | 1.0 | 0.963 (0.018) | 0.900 (0.026) | 1.007 (0.015) |
| | *2.5* | 1 | 1 | 1 |
| | 5.0 | 1.010 (0.003) | 1.021 (0.006) | 1.006 (0.003) |
| | 10.0 | 1.013 (0.004) | 1.026 (0.008) | 1.008 (0.004) |
| | none | 1.014 (0.005) | 1.027 (0.008) | 1.009 (0.005) |
| Conflict threshold $z$ | 2.0 | 1.038 (0.037) | 1.003 (0.003) | 1.020 (0.018) |
| | 2.5 | 1.005 (0.026) | 1.000 (0.000) | 1.006 (0.009) |
| | *3.0* | 1 | 1 | 1 |
| | 4.0 | 1.000 (0.000) | 1.000 (0.000) | 0.997 (0.005) |
| | 6.0 | 1.000 (0.000) | 1.000 (0.000) | 0.997 (0.005) |
| Stated anchor s.d. | 0.1 | 1.206 (0.075) | 1.051 (0.031) | 1.050 (0.024) |
| | *0.3* | 1 | 1 | 1 |
| | 0.6 | 1.055 (0.036) | 1.128 (0.048) | 1.018 (0.014) |
| | 1.0 | 1.093 (0.050) | 1.273 (0.079) | 1.031 (0.020) |
| | 2.0 | 1.117 (0.059) | 1.407 (0.113) | 1.038 (0.022) |
| Gossip rounds $T$ | 5 | 1.176 (0.046) | 1.074 (0.045) | 1.026 (0.022) |
| | 10 | 1.066 (0.019) | 1.019 (0.008) | 1.007 (0.014) |
| | 20 | 1.015 (0.008) | 1.005 (0.003) | 1.000 (0.005) |
| | 40 | 1.001 (0.002) | 1.001 (0.001) | 0.999 (0.001) |
| | *60* | 1 | 1 | 1 |
| | 120 | 1.000 (0.000) | 1.000 (0.000) | 1.000 (0.000) |
| Public sample size | 2000 | 1.001 (0.003) | 1.003 (0.002) | 1.003 (0.004) |
| | 5000 | 1.001 (0.003) | 1.002 (0.002) | 1.001 (0.003) |
| | *20000* | 1 | 1 | 1 |
| | 100000 | 1.001 (0.001) | 1.001 (0.001) | 1.001 (0.001) |

($\tau_S^2 = 0.5$), a rare minority, and disparities of either sign ($\mu_S = 0$, $\tau_S^2 = 0.5$). Besides GHB-lite, a hospital can decline to borrow its disparity: *GHB-lite, S local* uses the same release and the same network estimate for every other coefficient, but keeps the local prior for the coefficient of $S$; it costs no privacy. The comparators are FAIR-FATE [Salazar et al. 2023], implemented as its Algorithm 1 with the settings of its paper (Table 9), with statistical parity (SP) or equalized odds (EQO) as the fairness measure, and federated averaging [McMahan et al. 2017] under the same training protocol. Both train one global model with a server, and the server of FAIR-FATE [Salazar et al. 2023] holds a labelled validation set of a fifth of all patients. Neither is private. They are references for what a fairness-aware federated method does, not baselines at GHB-lite's budget.

For the minority group, GHB-lite or its variant is the most accurate private method in every scenario, and the most accurate of all methods whenever hospitals disagree about the disparity (Figure 13, Table 9). GHB-lite roughly halves the minority's excess log-loss of local estimation when the disparity is shared (17.8 against 34.3 $\times 10^{-3}$ nats) and when the minority is rare (25.0 against 42.4). When the minority is rare the non-private global model is better still (21.7); when the disparity is shared it is not (20.3). When disparities vary, a single model fits the minority badly (global 60.7,

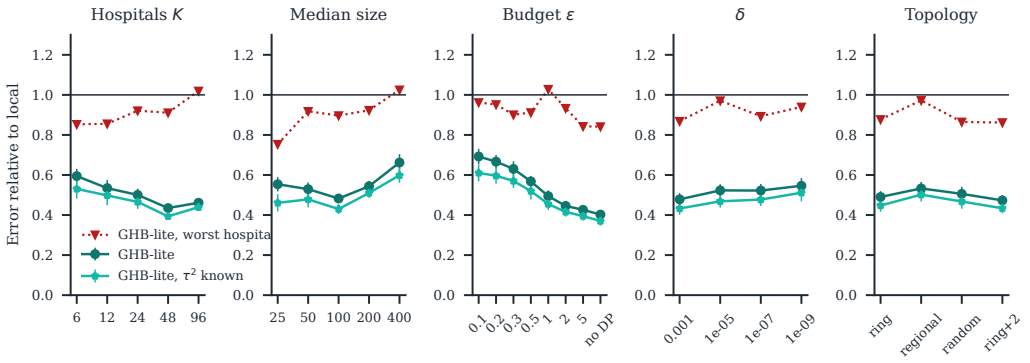



Fig. 12. Simulation 17, part B: GHB-lite across network conditions ($\tau^2 = 0.05$, $\epsilon = 1$, 24 hospitals of median size 100 on the ring with two-hop links unless varied; 40 replicates per point). Mean excess log-loss relative to local estimation for GHB-lite and for GHB-lite with $\tau^2$ known (±2 standard errors), and the largest ratio of any hospital's error to its own local error (dotted).

Table 8. Simulation 17, part B: exact values behind Figure 12. Mean excess log-loss of GHB-lite relative to local estimation (two standard errors in grey), the same for GHB-lite with $\tau^2$ known, the largest ratio of any hospital's error to its own local error, and the absolute mean excess log-loss of GHB-lite and of local estimation ($10^{-3}$ nats). $\tau^2 = 0.05$, $\epsilon = 1$, $\delta = 10^{-5}$, 24 hospitals of median size 100 on the ring with two-hop links unless varied.

| Condition | Value | GHB-lite / local | GHB-lite, $\tau^2$ known / local | Worst hospital | GHB-lite | Local |
|---|---|---|---|---|---|---|
| Hospitals $K$ | 6 | 0.595 (0.036) | 0.531 | 0.85 | 13.7 | 23.0 |
| | 12 | 0.534 (0.041) | 0.498 | 0.86 | 13.4 | 25.2 |
| | 24 | 0.500 (0.031) | 0.466 | 0.92 | 13.7 | 27.4 |
| | 48 | 0.435 (0.018) | 0.394 | 0.91 | 14.0 | 32.2 |
| | 96 | 0.461 (0.020) | 0.439 | 1.02 | 13.2 | 28.5 |
| Median size | 25 | 0.554 (0.036) | 0.461 | 0.75 | 33.9 | 61.2 |
| | 50 | 0.529 (0.033) | 0.478 | 0.92 | 23.0 | 43.5 |
| | 100 | 0.482 (0.022) | 0.430 | 0.90 | 13.3 | 27.6 |
| | 200 | 0.544 (0.026) | 0.510 | 0.92 | 8.7 | 16.1 |
| | 400 | 0.662 (0.042) | 0.599 | 1.03 | 5.6 | 8.4 |
| Budget $\epsilon$ | 0.1 | 0.692 (0.038) | 0.610 | 0.96 | 18.9 | 27.3 |
| | 0.2 | 0.667 (0.033) | 0.596 | 0.95 | 18.2 | 27.3 |
| | 0.3 | 0.630 (0.038) | 0.570 | 0.90 | 16.4 | 26.0 |
| | 0.5 | 0.568 (0.028) | 0.520 | 0.91 | 16.7 | 29.4 |
| | 1.0 | 0.495 (0.029) | 0.455 | 1.03 | 13.8 | 27.8 |
| | 2.0 | 0.446 (0.025) | 0.416 | 0.93 | 12.1 | 27.3 |
| | 5.0 | 0.426 (0.023) | 0.395 | 0.84 | 11.1 | 26.1 |
| | no DP | 0.402 (0.023) | 0.370 | 0.84 | 10.9 | 27.2 |
| $\delta$ | 0.001 | 0.478 (0.031) | 0.433 | 0.87 | 13.0 | 27.2 |
| | 1e-05 | 0.523 (0.031) | 0.468 | 0.97 | 14.5 | 27.8 |
| | 1e-07 | 0.522 (0.033) | 0.477 | 0.89 | 14.2 | 27.2 |
| | 1e-09 | 0.546 (0.038) | 0.511 | 0.94 | 14.9 | 27.2 |
| Topology | ring | 0.490 (0.029) | 0.448 | 0.88 | 13.9 | 28.4 |
| | regional | 0.533 (0.032) | 0.502 | 0.97 | 14.2 | 26.7 |
| | sparse random | 0.506 (0.036) | 0.467 | 0.86 | 13.6 | 27.0 |
| | ring, two-hop | 0.473 (0.025) | 0.434 | 0.86 | 13.1 | 27.7 |

FAIR-FATE [Salazar et al. 2023] 72.1 and 79.1), while GHB-lite reaches 27.3 and its variant 26.5. Whether GHB-lite imports other hospitals' disparity depends on that agreement. Because GHB-lite

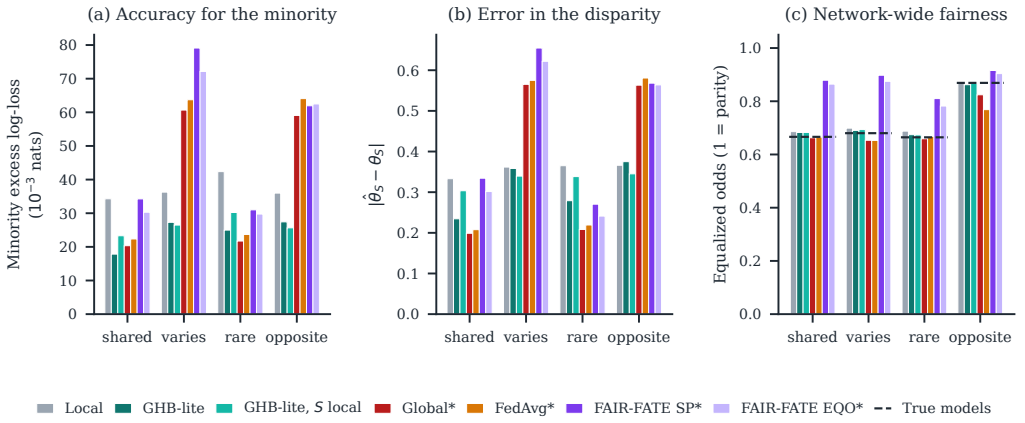



Fig. 13. Simulation 18: a protected attribute $S$ ($\epsilon = 1$, 24 hospitals, 100 replicates per scenario). (a) Mean excess log-loss of the minority group. (b) Mean absolute error of the estimated disparity, the coefficient of $S$. (c) Equalized odds of the network's decisions at threshold 0.5 (1 is parity); dashed: the true models. $*$: not private, with a server; FAIR-FATE [Salazar et al. 2023] also uses a labelled validation set at the server.

estimates $\tau$ coordinate by coordinate, a disparity on which hospitals agree is borrowed, and its error falls from 0.33 to 0.23 on the log-odds scale (from 0.37 to 0.28 with a rare minority). A disparity on which they disagree is barely borrowed, and its error stays at the local level (0.36 against 0.36; 0.38 against 0.37 with disparities of either sign, within Monte Carlo error). In those two scenarios, keeping the disparity local is slightly better (0.34 and 0.35), and in the other two it gives up most of the gain. So the option is a safeguard for networks that expect their disparities to differ, not a default. GHB-lite leaves the network's fairness ratios where the true models put them: equalized odds 0.68 against 0.67 for the true models when the disparity is shared, and within 0.02 of the truth in every scenario. FAIR-FATE [Salazar et al. 2023] does what it is designed to do and moves them toward parity (0.86 with $F$ = EQO). Here, however, the disparity is real, so parity is reached by distorting the predicted risks: the minority excess log-loss of FAIR-FATE is 30.3 against GHB-lite's 17.8 when the disparity is shared. This is the conflict that Section 2.3 anticipated. Calibrated, personalized risk and parity of decisions are different goals, and which one a clinical model should serve is a question for the deployment, not for the estimator.

## 7 REAL HOSPITAL NETWORKS

The simulations control everything; real data control nothing. This section asks whether GHB's advantages survive real hospitals, real heterogeneity and real sample sizes, against the federated and personalized methods that apply, at the same privacy budget. It reports two studies. The first is a confirmation on networks that no part of GHB's development had seen. Its rule, its predictions and the frozen configuration of every method were committed to the repository before it ran (Section 7.1). The second is the earlier benchmark of eight smaller networks, on which GHB-lite was developed and its limits were found (Section 7.2). The design of GHB in Section 5.2 came out of those limits, and every step of that development is reported, including the three pre-registered attempts that did not deliver: two failed their own criteria, and the third held its predictions but showed that its extensions did not help.

Table 9. Simulation 18: exact values behind Figure 13 (100 replicates per scenario). Mean excess log-loss over all patients, the minority ($S = 1$) and the majority ($10^{-3}$ nats; two standard errors in grey); mean absolute error of the estimated disparity $|\hat{\theta}_S - \theta_S|$; mean absolute calibration bias in the minority, $|\text{mean}(\hat{p} - p \mid S = 1)|$ (percentage points); the largest ratio of any hospital's excess log-loss to its local one (Worst); and the statistical parity, equal opportunity and equalized odds ratios of the network's decisions at threshold 0.5 (1 is parity). Bold: best of the private methods. †: not private, one model trained with a server. FAIR-FATE follows Algorithm 1 of [Salazar et al. 2023]: a third of the hospitals per round, 10 local epochs of mini-batch gradient descent (batches of 10, learning rate 0.01), $\lambda_0 = 0.5$, $\rho = 0.05$, MAX = 0.9, $\beta_0 = 0.9$, 100 rounds, and a labelled validation set of a fifth of all patients at the server; federated averaging [McMahan et al. 2017] uses the same protocol without the fairness update. All federated models start from the anchor $\mu_0$.

| | Excess log-loss | | | | | | Fairness of decisions | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | All | Minority | Majority | $\lvert\hat{\theta}_S - \theta_S\rvert$ | Calib. bias | Worst | SP | EO | EQO |
| *Shared disparity ($\tau_S^2 = 0.05$)* | | | | | | | | | |
| Local | 25.4 (0.8) | 34.3 (1.4) | 23.3 | 0.333 | 6.3 | 1.00 | 0.65 | 0.76 | 0.69 |
| GHB-lite | 13.2 (0.5) | **17.8** (0.8) | 11.7 | **0.234** | 4.9 | 0.86 | 0.65 | 0.76 | 0.68 |
| GHB-lite, $S$ local | 14.4 (0.5) | 23.3 (1.1) | 11.9 | 0.304 | 6.0 | 0.91 | 0.65 | 0.77 | 0.68 |
| Global† | 16.1 (0.5) | 20.3 (0.8) | 14.5 | 0.199 | 5.3 | 2.72 | 0.63 | 0.74 | 0.66 |
| FedAvg† [McMahan et al. 2017] | 17.7 (0.6) | 22.3 (1.1) | 16.0 | 0.208 | 5.5 | 2.56 | 0.63 | 0.74 | 0.67 |
| FAIR-FATE, SP† [Salazar et al. 2023] | 24.2 (1.3) | 34.3 (2.5) | 19.9 | 0.334 | 7.5 | 4.47 | 0.83 | 0.91 | 0.88 |
| FAIR-FATE, EQO† [Salazar et al. 2023] | 22.1 (0.9) | 30.3 (1.8) | 18.7 | 0.302 | 6.9 | 4.11 | 0.81 | 0.90 | 0.86 |
| True models | | | | | | | 0.63 | 0.75 | 0.67 |
| *Disparity varies ($\tau_S^2 = 0.5$)* | | | | | | | | | |
| Local | 25.9 (0.9) | 36.3 (1.7) | 23.7 | 0.362 | 6.4 | 1.00 | 0.65 | 0.74 | 0.70 |
| GHB-lite | 15.8 (0.6) | 27.3 (1.4) | 12.9 | 0.358 | 6.5 | 0.98 | 0.65 | 0.74 | 0.69 |
| GHB-lite, $S$ local | 15.5 (0.6) | **26.5** (1.4) | 12.7 | **0.340** | 6.3 | 0.96 | 0.65 | 0.74 | 0.69 |
| Global† | 28.7 (1.1) | 60.7 (2.8) | 15.3 | 0.565 | 11.7 | 4.83 | 0.62 | 0.75 | 0.65 |
| FedAvg† [McMahan et al. 2017] | 30.6 (1.2) | 63.7 (3.2) | 16.4 | 0.575 | 12.0 | 4.52 | 0.63 | 0.74 | 0.65 |
| FAIR-FATE, SP† [Salazar et al. 2023] | 37.1 (1.7) | 79.1 (4.8) | 18.7 | 0.655 | 13.4 | 7.62 | 0.89 | 0.92 | 0.90 |
| FAIR-FATE, EQO† [Salazar et al. 2023] | 34.5 (1.4) | 72.1 (3.7) | 18.3 | 0.622 | 12.7 | 7.03 | 0.84 | 0.92 | 0.87 |
| True models | | | | | | | 0.63 | 0.72 | 0.68 |
| *Rare minority (share 0.1)* | | | | | | | | | |
| Local | 24.1 (0.8) | 42.4 (1.8) | 22.7 | 0.365 | 7.8 | 1.00 | 0.65 | 0.77 | 0.69 |
| GHB-lite | 13.0 (0.5) | **25.0** (2.0) | 11.9 | **0.279** | 6.2 | 0.84 | 0.64 | 0.76 | 0.67 |
| GHB-lite, $S$ local | 13.5 (0.5) | 30.2 (1.6) | 12.0 | 0.338 | 7.3 | 0.88 | 0.64 | 0.76 | 0.67 |
| Global† | 15.8 (0.5) | 21.7 (1.0) | 15.1 | 0.208 | 5.4 | 2.37 | 0.62 | 0.74 | 0.66 |
| FedAvg† [McMahan et al. 2017] | 17.3 (0.7) | 23.7 (1.2) | 16.5 | 0.219 | 5.6 | 2.30 | 0.63 | 0.75 | 0.67 |
| FAIR-FATE, SP† [Salazar et al. 2023] | 23.3 (1.6) | 31.0 (2.5) | 22.3 | 0.270 | 6.4 | 4.10 | 0.76 | 0.86 | 0.81 |
| FAIR-FATE, EQO† [Salazar et al. 2023] | 23.0 (1.5) | 29.8 (2.4) | 22.2 | 0.241 | 6.2 | 3.95 | 0.74 | 0.84 | 0.78 |
| True models | | | | | | | 0.63 | 0.75 | 0.66 |
| *Opposite disparities ($\mu_S = 0$, $\tau_S^2 = 0.5$)* | | | | | | | | | |
| Local | 26.1 (0.9) | 36.0 (1.8) | 24.1 | 0.366 | 6.4 | 1.00 | 0.87 | 0.89 | 0.87 |
| GHB-lite | 16.0 (0.7) | 27.4 (2.6) | 13.2 | 0.375 | 6.5 | 0.94 | 0.86 | 0.89 | 0.86 |
| GHB-lite, $S$ local | 15.3 (0.5) | **25.7** (1.5) | 12.7 | **0.345** | 6.2 | 0.90 | 0.86 | 0.89 | 0.87 |
| Global† | 28.5 (1.0) | 59.1 (2.8) | 15.7 | 0.563 | 11.4 | 4.09 | 0.82 | 0.86 | 0.82 |
| FedAvg† [McMahan et al. 2017] | 31.4 (1.2) | 64.1 (3.2) | 17.5 | 0.581 | 11.8 | 4.07 | 0.77 | 0.81 | 0.77 |
| FAIR-FATE, SP† [Salazar et al. 2023] | 31.3 (1.3) | 61.9 (2.8) | 17.8 | 0.568 | 11.5 | 5.76 | 0.92 | 0.93 | 0.92 |
| FAIR-FATE, EQO† [Salazar et al. 2023] | 32.4 (1.4) | 62.5 (2.7) | 19.0 | 0.564 | 11.5 | 5.86 | 0.91 | 0.93 | 0.90 |
| True models | | | | | | | 0.86 | 0.88 | 0.87 |

## 7.1 A pre-registered test on unseen networks

*Networks and model.* The Cancer Genome Atlas [Cancer Genome Atlas Research Network et al. 2013; Grossman et al. 2016] records, for every patient, the institution that contributed the tissue. Each cancer type is a network, and each institution with at least five eligible patients is a hospital. The outcome is death within two years of diagnosis; patients alive with less than two years of follow-up are excluded. Every hospital fits a logistic regression with eight coefficients: intercept, age,

sex, pathologic stage III, stage IV, tumour stage T3 or T4, nodal involvement and a prior malignancy, with $\|x\| \leq \sqrt{8}$ by construction (FIGO stage replaces the AJCC stage for the gynecologic cancers). Twelve networks served for development: lung squamous and adenocarcinoma, head and neck, bladder, stomach, liver, clear-cell and papillary kidney, colon, rectum, pancreas and oesophagus (5 to 17 hospitals each). The confirmation set was then fixed by a rule on counts alone, before any model was fitted to it. It contains every further TCGA project with at least four such hospitals and at least fifteen deaths: breast (8 hospitals), mesothelioma (9), ovarian (15), endometrial (15) and uterine carcinosarcoma (5). Hospitals hold 5 to 109 eligible patients.

*Protocol.* Every hospital is evaluated on its own held-out patients by ten repetitions of stratified five-fold cross-validation, and errors are held-out log-loss relative to local estimation under the default prior $\mathcal{N}(0, 2.5^2)$. Every private method spends the same budget, $\epsilon \in \{1, 10\}$ with $\delta = 10^{-5}$ under zero-concentrated accounting. The literature methods are DP-FedAvg [McMahan et al. 2018] with local fine-tuning, with and without secure aggregation [Bonawitz et al. 2017]; clustered DP-FedAvg [Ghosh et al. 2020]; APFL [Deng et al. 2020]; Ditto [Li et al. 2021a]; and pFedMe [T. Dinh et al. 2020]. Each has a grid of 12 to 54 configurations, and how it is tuned decides how it compares. In the *realistic* arm, each method uses the configuration with the lowest mean error over the twelve development networks, frozen before the confirmation. This is what a consortium could do with networks it already has. In the *oracle* arm, each method uses the configuration that is best on the confirmation network's own held-out folds, which no hospital can do. GHB is not tuned: its configuration (Section 5.2) was chosen among about fifteen variants on the development networks and frozen together with the baselines'. The two privacy budgets bracket the regime of interest. At $\epsilon = 1$ the joint releases of a network of a few hundred patients are noisy. At $\epsilon = 10$ they are accurate, and borrowing can pay.

*How GHB was developed.* GHB-lite, with an evidence-weighted default prior, was the starting point. On the twelve development networks it is no better than a local model with the same prior at $\epsilon = 1$ (0.927 against 0.927), and worse than it at $\epsilon = 10$ (0.941 against 0.927). A pre-registered extension of GHB-lite added four components: a no-borrowing option, a network-learned prior scale, correlated heterogeneity and borrowing across cancer types. It was confirmed on six networks that were then unseen. Its predictions held, but at $\epsilon = 10$ every variant was worse than local estimation, which the extension was meant to prevent; that confirmation made those six networks part of the development set. The diagnosis was the bias and the noise of GHB-lite's per-hospital releases (Section 5.2), and GHB replaces them by the two joint releases. On the development networks GHB reaches 0.926 at $\epsilon = 1$ and 0.904 at $\epsilon = 10$, against 0.934 and 0.913 for the best literature method tuned on the same networks. The ablation in the supplement removes one component at a time. Borrowing across cancer types and propagating the privacy covariance matter most, at both budgets (without them the mean error rises by 0.007 to 0.035). Each of the other components changes it by less than 0.007.

*Predictions and outcome.* The pre-registration made three predictions, and all three held (Table 10).

- *GHB has a lower mean error than every literature method in the realistic arm, at both budgets.* At $\epsilon = 1$, GHB reaches 0.928, against 0.976 for the best of them, DP-FedAvg [McMahan et al. 2018] with fine-tuning, and between 1.000 and 1.303 for the others. At $\epsilon = 10$, GHB reaches 0.915, against 0.919 for Ditto [Li et al. 2021a] and 0.927 for pFedMe [T. Dinh et al. 2020]; the other four reach between 0.960 and 0.990.
- *At $\epsilon = 10$ GHB improves on GHB-lite.* It does, from 0.965 to 0.915.
- *At $\epsilon = 10$ GHB beats the best realistic method in at least three of the five networks.* It beats Ditto [Li et al. 2021a] in exactly 3, and every other method in at least four.

Table 10. Pre-registered confirmation on five TCGA networks that no part of the development had seen (hospitals: BRCA 8, MESO 9, OV 15, UCEC 15, UCS 5): held-out log-loss relative to local estimation under the default prior (ten repetitions of five-fold cross-validation). GHB and GHB-lite have nothing to tune. Every literature method uses the configuration that is best on the twelve development networks, frozen before this run. Bold: the lowest per network among methods that use other hospitals, oracle excluded. Wins: networks in which GHB is more accurate. ‡: no other hospital. ⋆: for each network, the best configuration of the best of the six methods on that network's own held-out folds, which no hospital can do.

| Method | BRCA | MESO | OV | UCEC | UCS | Mean | Wins |
|---|---|---|---|---|---|---|---|
| $\epsilon = 1$ | | | | | | | |
| **GHB**, no tuning | **0.896** | 0.895 | 0.979 | **0.982** | **0.888** | 0.928 | |
| GHB-lite (evidence prior), no tuning | 0.901 | **0.893** | 0.982 | 0.988 | 0.895 | 0.932 | 4/5 |
| Local, evidence prior‡ | 0.902 | 0.895 | 0.983 | 0.992 | 0.894 | 0.933 | 5/5 |
| DP-FedAvg + FT, tuned on development [McMahan et al. 2018] | 0.898 | 0.952 | 1.004 | 1.108 | 0.921 | 0.976 | 5/5 |
| DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018] | 0.911 | 1.027 | 0.999 | 1.070 | 1.019 | 1.005 | 5/5 |
| APFL, tuned on development [Deng et al. 2020] | 0.918 | 0.987 | 1.010 | 1.134 | 0.986 | 1.007 | 5/5 |
| Ditto, tuned on development [Li et al. 2021a] | 0.967 | 1.503 | 0.978 | 0.997 | 1.556 | 1.200 | 4/5 |
| pFedMe, tuned on development [T. Dinh et al. 2020] | 1.029 | 1.832 | **0.977** | 1.013 | 1.661 | 1.303 | 4/5 |
| Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018] | 0.926 | 0.989 | 1.007 | 1.112 | 0.965 | 1.000 | 5/5 |
| Best of the six, oracle per network⋆ | 0.895 | 0.914 | 0.974 | 0.962 | 0.917 | 0.933 | 2/5 |
| $\epsilon = 10$ | | | | | | | |
| **GHB**, no tuning | 0.868 | 0.900 | **0.967** | 0.965 | 0.873 | 0.915 | |
| GHB-lite (evidence prior), no tuning | 0.948 | 0.958 | 1.005 | 1.005 | 0.907 | 0.965 | 5/5 |
| Local, evidence prior‡ | 0.902 | 0.895 | 0.983 | 0.992 | 0.894 | 0.933 | 4/5 |
| DP-FedAvg + FT, tuned on development [McMahan et al. 2018] | 0.889 | 0.947 | 0.997 | 1.061 | 0.906 | 0.960 | 5/5 |
| DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018] | 0.903 | 1.008 | 0.975 | **0.957** | 1.109 | 0.990 | 4/5 |
| APFL, tuned on development [Deng et al. 2020] | 0.893 | 0.966 | 0.994 | 1.016 | 0.960 | 0.966 | 5/5 |
| Ditto, tuned on development [Li et al. 2021a] | 0.878 | **0.893** | 0.974 | 0.982 | **0.866** | 0.919 | 3/5 |
| pFedMe, tuned on development [T. Dinh et al. 2020] | **0.856** | 0.926 | 0.969 | 0.987 | 0.898 | 0.927 | 4/5 |
| Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018] | 0.894 | 0.940 | 1.002 | 1.095 | 0.894 | 0.965 | 5/5 |
| Best of the six, oracle per network⋆ | 0.845 | 0.877 | 0.962 | 0.900 | 0.864 | 0.890 | 0/5 |

*Against the oracle.* GHB was also more accurate than local estimation at both budgets, which GHB-lite was not. Against the oracle arm, which the pre-registration made no prediction about, the picture depends on the budget. At $\epsilon = 1$, GHB has a lower mean error than every one of the six methods even when each is tuned on the test folds (0.939 for the best of them). Only the best method chosen afresh for each network comes close (0.933). At $\epsilon = 10$, GHB ties DP-FedAvg [McMahan et al. 2018] with fine-tuning tuned on the test folds (0.915), and is slightly behind Ditto [Li et al. 2021a] (0.911) and DP-FedAvg [McMahan et al. 2018] with secure aggregation (0.912). It is clearly behind pFedMe [T. Dinh et al. 2020] (0.898), and behind the best method chosen afresh for every network (0.890). These are the margins that tuning on a hospital's own test patients buys. A deployment cannot buy them.

*Tuned against tuned.* GHB has nothing to tune, which leaves a question: is its lead only the price the literature methods pay for being tuned on other networks? To answer it, after the confirmation and outside the pre-registration, GHB was given a grid of 36 configurations around its frozen one. The grid varies the scale of the half-Cauchy prior on $\tau$ (0.1 to 1), the weight of the non-borrowing components (none to equal), and the split of the budget between the two releases ($\frac{1}{4}$ to $\frac{3}{4}$). GHB was then tuned exactly as the baselines were: on the twelve development networks, and on each confirmation network's own held-out folds (Table 11). This run draws new privacy noise, so the frozen configuration reaches 0.933 and 0.911 rather than 0.928 and 0.915, a reminder that differences of a few thousandths are within the noise of a single run. Tuning changes GHB little. Tuned on development it reaches 0.933 and 0.907, and tuned on the test folds 0.931 and 0.904. With the same tuning on both sides, GHB is more accurate than every literature method at $\epsilon = 1$, under development tuning and under hindsight tuning alike. At $\epsilon = 10$ it is more accurate than every method under development tuning, and under hindsight tuning it is more accurate than five of the

Table 11. Tuned against tuned, on the five confirmation networks (exploratory, after the pre-registered run): mean held-out log-loss relative to local estimation. GHB tuned over 36 configurations, every literature method over its grid of 12 to 54. Development: the configuration with the lowest mean error over the twelve development networks, applied unchanged. Oracle: the best configuration on each network's own held-out folds. GHB's rows come from a rerun with new privacy noise.

| | $\epsilon = 1$ | | $\epsilon = 10$ | |
|---|---|---|---|---|
| Method | Development | Oracle | Development | Oracle |
| GHB, frozen configuration | 0.933 | 0.933 | 0.911 | 0.911 |
| **GHB**, tuned like the others | **0.933** | **0.931** | **0.907** | 0.904 |
| DP-FedAvg + FT [McMahan et al. 2018] | 0.976 | 0.939 | 0.960 | 0.915 |
| DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018] | 1.005 | 0.983 | 0.990 | 0.912 |
| APFL [Deng et al. 2020] | 1.007 | 0.979 | 0.966 | 0.919 |
| Ditto [Li et al. 2021a] | 1.200 | 1.196 | 0.919 | 0.911 |
| pFedMe [T. Dinh et al. 2020] | 1.303 | 1.241 | 0.927 | **0.898** |
| Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] | 1.000 | 0.970 | 0.965 | 0.930 |

six. The sixth, pFedMe [T. Dinh et al. 2020], reaches 0.898 and beats it in three of the five networks. GHB's frozen configuration is already close to the best it can reach, which is what a method without tuning parameters needs.

*What the confirmation supports.* Five networks are a small sample. At $\epsilon = 10$ the margin over Ditto [Li et al. 2021a] is small enough, 0.919 against 0.915 with three wins in five, that a different set of networks could reverse it. The claim the confirmation supports is therefore specific. Without any tuning, on networks it had not seen, GHB was more accurate than every federated and personalized method tuned in the way a consortium could tune it, at both budgets. Under strong privacy, it was more accurate than every one of them even with tuning that no consortium has. Given the same tuning as the others, it keeps that lead, except against pFedMe [T. Dinh et al. 2020] tuned with hindsight under weak privacy. It does not support the claim that GHB dominates the literature in every network or under every way of tuning.

## 7.2 Eight smaller networks

GHB-lite was developed on the eight networks below, and its limits were found on them. GHB was added afterwards, with its configuration frozen as in Section 7.1 and nothing else changed, so every other number is as in the runs without it. Two of these networks, Arizona Medicare and UCI Heart Disease, played no part in GHB's development. The six TCGA networks here use a simpler head than in Section 7.1, and each network is fitted on its own, so GHB cannot borrow across cancer types.

*Eight real networks.* All data are public and need no credentials. *Arizona Medicare* [Hilbe 2011]: 1,466 admissions for percutaneous cardiovascular procedures at 43 hospitals with at least five patients (median 29), in-hospital death (34%), predictors age 80 or over, HMO membership, urgent and emergency admission. *The Cancer Genome Atlas* [Cancer Genome Atlas Research Network et al. 2013; Grossman et al. 2016]: six networks, one per cancer type (lung squamous, head and neck, bladder, stomach, liver and clear-cell kidney carcinoma), each hospital the tissue source site that contributed its patients (7 to 17 hospitals, median 11 to 29 patients), death within two years of diagnosis, predictors age, sex, and pathologic stage III and IV. *UCI Heart Disease* [Detrano et al. 1989]: the four hospitals described below, used as the development network. With four or five coefficients and a few dozen patients per hospital these networks hold little to share, which is the setting of siloed smallness and also the hardest one for any collaborative method.

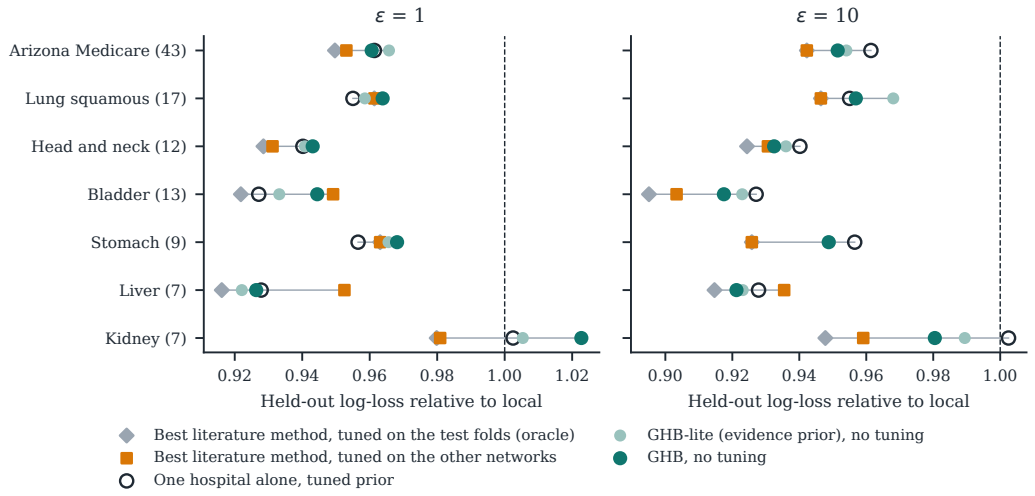



Fig. 14. Seven real networks (hospitals in parentheses): held-out log-loss relative to local estimation, at $\epsilon = 1$ and $\epsilon = 10$. Dark teal: GHB; light teal: GHB-lite with the evidence-weighted prior; neither is tuned. Orange: the best of twelve federated and personalized methods for that network, each tuned on the other seven networks. Grey: the same, tuned on the network's own held-out folds (oracle). Open circle: one hospital alone with a tuned prior. Choosing the best of twelve methods separately for every network is itself an advantage no hospital has.

*Protocol.* Every hospital is evaluated on its own held-out patients by repeated stratified five-fold cross-validation (100 splits per network), and errors are reported as held-out log-loss relative to local estimation with the default prior. Every private method spends the same budget per hospital, $\epsilon \in \{1, 10\}$ and $\delta = 10^{-5}$, with zero-concentrated accounting. GHB and GHB-lite run without an anchor, because no published model is assumed; GHB-lite inverts its releases with a nominal public feature distribution fixed before the data were seen, which GHB does not need. Seven networks are evaluated: Arizona Medicare and the six TCGA networks. UCI Heart Disease, the eighth, served for development and enters only the tuning of the baselines. Twelve literature methods are compared: DP-FedAvg [McMahan et al. 2018] alone; DP-FedAvg with local fine-tuning, with and without secure aggregation [Bonawitz et al. 2017]; decentralized DP-SGD [Lian et al. 2017]; clustered DP-FedAvg [Ghosh et al. 2020]; a one-shot Ditto-style model [Li et al. 2021a]; and the personalized methods Ditto [Li et al. 2021a], pFedMe [T. Dinh et al. 2020], APFL [Deng et al. 2020], Per-FedAvg [Fallah et al. 2020], FedRep [Collins et al. 2021] and FedProx [Li et al. 2020], each with private gradients under the same accounting (SCAFFOLD [Karimireddy et al. 2020] and FedProx with one full-batch local step coincide with DP-FedAvg). Each has a grid of hyperparameters, and how it is tuned decides how it compares. Three tuning arms are reported. *Oracle*: tuned on each network's own held-out folds, an upper bound no hospital can reach. *Leave-one-network-out*: the configuration with the lowest mean error on the other seven networks, the most generous tuning a hospital could actually do. *Development*: tuned on UCI Heart Disease alone and frozen. GHB and GHB-lite with the evidence-weighted prior (GHB-lite-EB) have nothing to tune. Two local-only references separate what collaboration contributes from what regularization does: local estimation with a tuned prior scale, and local gradient descent with a tuned number of steps.

Table 12. Seven real networks: held-out log-loss relative to local estimation, mean over the networks, for every method in the oracle arm (tuned on the network's own held-out folds) and the leave-one-network-out arm (tuned on the other networks); Wins: in how many of the seven networks GHB, untuned, has a lower error than that method in the leave-one-network-out arm. ‡: no other hospital. Every method spends the same budget. Per-network values are in the supplement.

| | $\epsilon = 1$ | | | $\epsilon = 10$ | | |
|---|---|---|---|---|---|---|
| Method | Oracle | Other networks | Wins | Oracle | Other networks | Wins |
| *GHB and the local reference* | | | | | | |
| **GHB**, no tuning | 0.961 | 0.961 | | 0.944 | 0.944 | |
| GHB-lite, no tuning | 1.000 | 1.000 | | 0.964 | 0.964 | |
| GHB-lite, evidence prior, no tuning | 0.956 | 0.956 | | 0.949 | 0.949 | |
| GHB-lite with secure aggregation, no tuning | 0.982 | 0.982 | | 0.950 | 0.950 | |
| GHB-lite, tuned prior | 0.951 | **0.953** | | 0.948 | 0.950 | |
| Local, tuned prior‡ | 0.951 | 0.953 | | 0.951 | 0.953 | |
| *Federated and personalized methods (same budget)* | | | | | | |
| Ditto [Li et al. 2021a] | 0.962 | 0.962 | 4 | 0.944 | 0.949 | 5 |
| pFedMe [T. Dinh et al. 2020] | 0.970 | 0.971 | 3 | 0.939 | **0.942** | 3 |
| DP-FedAvg + FT, sec. agg. [McMahan et al. 2018] | **0.949** | 0.985 | 3 | **0.928** | 0.946 | 3 |
| APFL [Deng et al. 2020] | 0.958 | 0.988 | 6 | 0.944 | 0.963 | 6 |
| DP-FedAvg + FT [McMahan et al. 2018] | 0.951 | 0.994 | 4 | 0.940 | 0.952 | 4 |
| Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] | 0.954 | 0.997 | 5 | 0.948 | 0.967 | 6 |
| Decentralized DP-SGD [Lian et al. 2017] | 0.972 | 1.009 | 7 | 0.947 | 0.964 | 6 |
| FedProx [Li et al. 2020] | 1.006 | 1.013 | 6 | 0.942 | 0.958 | 5 |
| FedRep [Collins et al. 2021] | 1.045 | 1.046 | 7 | 1.013 | 1.023 | 6 |
| DP-FedAvg [McMahan et al. 2018] | 1.085 | 1.112 | 7 | 1.029 | 1.044 | 7 |
| Per-FedAvg [Fallah et al. 2020] | 1.161 | 1.165 | 7 | 1.009 | 1.025 | 7 |
| Ditto-style (one-shot) [Li et al. 2021a] | 1.321 | 1.333 | 7 | 0.984 | 1.015 | 7 |

*With strong privacy, collaboration adds little for anyone.* At $\epsilon = 1$ (Figure 14, Table 12) GHB reduces the error of local estimation to 0.961 on average (0.926 to 1.023 across networks). Tuned on the other networks, no literature method does better on average. Ditto [Li et al. 2021a] comes closest (0.962), and GHB beats each of the twelve methods in at least three of the seven networks. Tuned on each network's own test folds, DP-FedAvg [McMahan et al. 2018] with fine-tuning and secure aggregation reaches 0.949. But at this budget a single hospital that regularizes well does as well as any collaborative method: local estimation with a tuned prior reaches 0.953, and GHB-lite-EB 0.956. These networks hold a few hundred patients each (Medicare about 1,500), and noise calibrated to that total still swamps what a head of four or five coefficients could borrow. The privacy noise of each per-hospital release is many times the hospital's own sampling error (Section 7.2.1 below), so GHB-lite falls back to regularization.

*With weaker privacy, GHB is level with the best tuned methods.* At $\epsilon = 10$, collaboration starts to pay. GHB (0.944) is more accurate than local estimation with a tuned prior in 6 of the seven networks. Tuned on the other networks, only pFedMe [T. Dinh et al. 2020] is more accurate on average (0.942); DP-FedAvg [McMahan et al. 2018] with fine-tuning and secure aggregation reaches 0.946, Ditto [Li et al. 2021a] 0.949, and the other nine methods are less accurate. Tuned on the test folds, DP-FedAvg [McMahan et al. 2018] with fine-tuning and secure aggregation leads by about two percent (0.928). GHB's worst hospital is the liver network's hospital whose 22 patients all died, where every method that borrows fails (1.53 for GHB). The gain over GHB-lite-EB is small here (0.949), unlike in Section 7.1. These networks are fitted one at a time, so GHB cannot borrow across cancer types, which the ablation found to be one of its two most important components.

*Tuning is most of the baselines' advantage.* The comparison turns on how the literature methods are tuned, and GHB has nothing to tune. Every baseline loses accuracy when its settings must come

from other networks, and several become unstable. Tuned on UCI Heart Disease alone, the mean error at $\epsilon = 1$ becomes 1.022 for DP-FedAvg [McMahan et al. 2018] with fine-tuning and secure aggregation, 1.045 for Ditto [Li et al. 2021a] and 1.054 for pFedMe [T. Dinh et al. 2020], against 0.961 for GHB, and FedRep [Collins et al. 2021] diverges. A small hospital cannot tune on its own test patients, and has few patients to hold out for validation. The leave-one-network-out arm is the most a consortium could do for it.

*The road to GHB.* The first real-data run placed GHB-lite at the local level (1.000 at $\epsilon = 1$): its fixed default prior $\mathcal{N}(0, 2.5^2)$ is too loose for hospitals of a few dozen patients. Two improvements were then pre-registered on these networks, and both failed their criteria; the supplement gives the full record. Integrating the scale of the default prior out by each hospital's evidence (GHB-lite-EB) improved on GHB-lite in 6 of the seven networks, but raised the worst-hospital ratio in the liver network from 1.56 to 1.86. Its gain equals that of a local model with the same prior, so it comes from regularization, not from borrowing. Secure aggregation of per-hospital quantities, still inverted under a public feature distribution, improved on GHB-lite-EB in only 1 of seven. The lesson that led to GHB is in what neither changed: as long as the network estimate is read through a public feature distribution and per-hospital noise, it carries too little information to beat a well-regularized hospital.

*7.2.1 How much the size of a hospital matters.* Why per-hospital releases carry so little is clearest in the smallest network, UCI Heart Disease, where the size of each hospital can be varied by subsampling. This analysis concerns GHB-lite and the baselines it was first compared with, tuned on the held-out folds; it explains the limit that GHB was built to remove. The UCI Heart Disease data [Detrano et al. 1989] are the classic multi-site benchmark of federated learning in healthcare [Ogier du Terrail et al. 2022]. They hold the records of patients referred for coronary angiography at four hospitals: the Cleveland Clinic (303 patients), the Hungarian Institute of Cardiology in Budapest (294), the University Hospitals of Zurich and Basel (123) and the VA Medical Center in Long Beach (200). The prevalence of angiographic coronary disease ranges from 36% in Budapest to 93% in Zurich and Basel, so the heterogeneity is real and large. Each hospital fits the pretest-probability model of coronary disease, a logistic regression on age, sex and chest-pain type (asymptomatic, anginal, or non-anginal as reference), the predictors that every site records completely. Age is scaled by public constants, $(\text{age} - 55)/25$ clipped to $[-1, 1]$, so that $\|x\| \le \sqrt{5}$ without fitting anything to the data. No published model is assumed, so GHB-lite runs without an anchor: it releases the score at the public point 0, linearizes under the default prior, and gossips on a ring of four hospitals. The public feature distribution needed for the inversion is a nominal chest-pain clinic population (age $\mathcal{N}(55, 9^2)$, 75% men, 50% asymptomatic, 30% anginal pain), not the data, so every hospital is under a real case-mix shift. Evaluation is by repeated stratified five-fold cross-validation within each hospital (100 splits). Each hospital trains on its training folds, subsampled to at most 20, 40 or 80 patients to emulate siloed smallness, or on all of them, and is scored by log-loss on its own held-out fold. The baselines are the same as in Simulation 15, with their hyperparameters chosen on the held-out folds themselves, which favours them. A local-only control, gradient descent on the hospital's own data with a tuned number of steps, separates what they gain by collaborating from what they gain by regularizing.

The result is mixed, and it is consistent with the theory. At $\epsilon = 1$, no private method gains from collaboration in this network. GHB-lite stays within 1% of local estimation at every training size (1.00 with at most 20 patients, 1.00 with all of them), and its worst hospital stays within a factor 1.03 of its local error, at $\epsilon = 5$ too. The privacy noise explains why. On the parameter scale, a hospital of 40 patients releases noise with a standard deviation about 26 times its own sampling

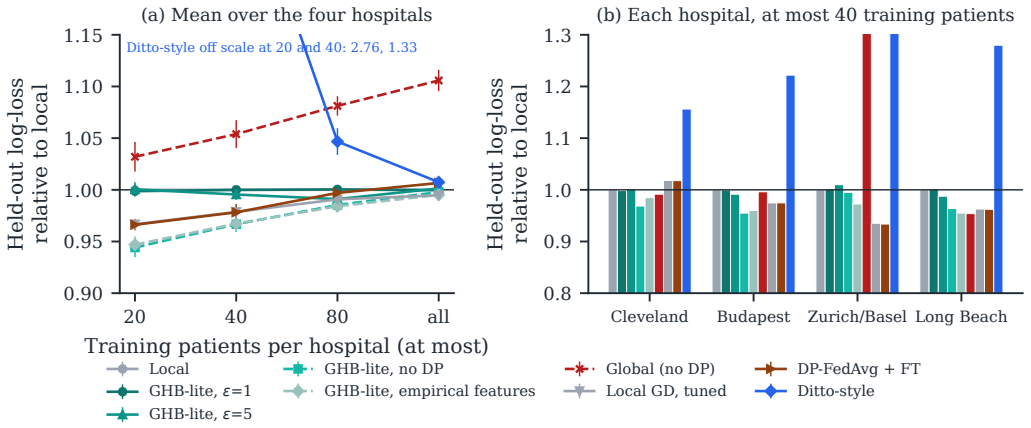



Fig. 15. UCI Heart Disease, four hospitals: held-out log-loss relative to local estimation (repeated stratified five-fold cross-validation, 100 splits; ±2 paired standard errors). (a) Mean over hospitals against the number of training patients per hospital. (b) Each hospital, with at most 40 training patients (bars clipped at 1.3; values in Table 13 and the supplement). Dashed: without privacy. The baselines' hyperparameters are tuned on the held-out folds.

error at $\epsilon = 1$, and about 10 times at 300 patients. With only three other hospitals to average over, $M$ in Equation (4) dwarfs $\tau^2$ and Equation (3) leaves almost nothing to borrow. GHB-lite's posterior over $\tau$ and the hospital's own evidence detect this and fall back to local estimation, without any tuning. The tuned DP-FedAvg [McMahan et al. 2018] with fine-tuning appears to gain (0.97 with 20 patients). The local-only control gains exactly as much (0.97), because the chosen configuration is a single private round followed by 30 local steps, so its gain is early stopping tuned on the test folds, not borrowing. The Ditto-style [Li et al. 2021a] model, pulled toward the average of the private releases, fails badly with small hospitals (2.76). Without privacy, GHB-lite borrows: 0.94 of the local error with at most 20 training patients, 0.97 with 40 and 0.99 with 80, fading to 1.00 with all of them, as Equation (3) predicts when $n_k$ grows. The nominal public feature distribution costs nothing: replacing it by the pooled empirical features gives 0.95 with 20 patients. A single pooled model, also without privacy, is worse than local estimation on average at every size (1.03 to 1.11) and makes the Swiss hospital, with its 93% prevalence, 1.57 to 1.81 times worse off. GHB-lite without privacy keeps every hospital within 2% of its local error. The lesson for deployment is the one the theory states. In a network of four hospitals of tens to a few hundred patients, a guarantee of $\epsilon = 1$ per release leaves nothing to share, and GHB-lite's value there is that it does no harm. Collaboration under privacy needs more hospitals (Simulation 17), releases that are calibrated to the whole network rather than to each hospital, which is what GHB does (Section 5.2), or a larger budget.

### 7.3 What the real data say

Four findings hold across the real networks we report. First, under strong privacy, per-hospital releases carry too little to borrow: GHB-lite, however regularized, does no better than a hospital alone, and a method that releases per-hospital quantities cannot beat that floor by much. Second, releasing only network sums, calibrated to the whole network and read through the hospitals' real information rather than a public feature distribution, is what lets collaboration pay. On the five unseen cancer networks, GHB was more accurate than every literature method tuned as a

Table 13. UCI Heart Disease, GHB-lite: held-out log-loss relative to local estimation (two paired standard errors in grey) and the largest ratio of any hospital's held-out log-loss to its local one (Worst), by the number of training patients per hospital (100 splits). First row: local log-loss (nats). DP-FedAvg [McMahan et al. 2018] + FT, Ditto-style [Li et al. 2021a] and local GD are tuned on the held-out folds. †: not private.

| | ≤ 20 patients | | ≤ 40 | | ≤ 80 | | All | |
|---|---|---|---|---|---|---|---|---|
| Method | Relative | Worst | Relative | Worst | Relative | Worst | Relative | Worst |
| Local log-loss | 0.483 | | 0.460 | | 0.443 | | 0.434 | |
| Local | 1.000 (0.000) | 1.00 | 1.000 (0.000) | 1.00 | 1.000 (0.000) | 1.00 | 1.000 (0.000) | 1.00 |
| GHB-lite, $\epsilon = 1$ | 0.999 (0.001) | 1.00 | 1.000 (0.002) | **1.00** | 1.000 (0.002) | 1.01 | **1.000** (0.002) | **1.00** |
| GHB-lite, $\epsilon = 5$ | 1.000 (0.007) | 1.01 | 0.995 (0.005) | 1.01 | **0.991** (0.004) | **1.00** | 1.001 (0.004) | 1.03 |
| DP-FedAvg + FT, $\epsilon = 1$ [McMahan et al. 2018] | **0.966** (0.005) | **0.99** | **0.978** (0.008) | 1.02 | 0.997 (0.006) | 1.03 | 1.007 (0.005) | 1.05 |
| Ditto-style, $\epsilon = 1$ [Li et al. 2021a] | 2.765 (0.334) | 5.12 | 1.327 (0.054) | 2.06 | 1.047 (0.013) | 1.16 | 1.007 (0.004) | 1.04 |
| Local GD, tuned stopping | 0.967 (0.005) | 0.99 | 0.979 (0.008) | 1.02 | 0.991 (0.002) | 1.00 | 0.995 (0.002) | 1.00 |
| GHB-lite, no DP† | 0.944 (0.009) | 1.02 | 0.967 (0.007) | 1.00 | 0.985 (0.005) | 1.01 | 0.998 (0.004) | 1.02 |
| GHB-lite, empirical features† | 0.947 (0.008) | 0.99 | 0.967 (0.007) | 0.98 | 0.984 (0.004) | 1.00 | 0.995 (0.002) | 1.00 |
| Global† | 1.032 (0.014) | 1.58 | 1.054 (0.013) | 1.57 | 1.081 (0.009) | 1.59 | 1.106 (0.010) | 1.81 |

consortium could tune it, at both budgets, and under strong privacy more accurate than each even when tuned on the test data. Third, the gain grows with what the network can share: it is largest with the richer head and several networks fitted together, and small on networks of a few hundred patients with a four-coefficient head, where GHB is level with the best tuned methods rather than ahead. Fourth, tuning is most of what separates the literature methods from GHB: tuned on the test data, which no hospital can do, the best of them lead by one to two percent under weak privacy. The real data do not show GHB dominating the literature everywhere, and the paper does not claim it.

## 8 DISCUSSION

The simulations and the real networks answer the question of Section 3 from two sides: the first under assumptions that GHB-lite was built for, the second under conditions that nobody controlled. This section draws the practical consequences: when each protocol should be used (Section 8.1), what the method does not protect against and what the evidence does not show (Section 8.2), and the problems that remain open (Section 8.3).

### 8.1 When to use GHB, and when not

The simulations and the real networks support a simple reading of the theory. Collaboration pays when hospitals are small relative to their task and differ moderately: Equation (3) then adds tens of virtual patients to every hospital, which matters to small ones. When hospitals differ strongly, no method gains much, and integrating $\tau$ out keeps GHB close to local estimation rather than worse than it. When hospitals barely differ and a trusted environment for pooling exists, a single pooled model is better, and GHB can run inside that environment without privacy noise (Section 3).

Which protocol to run depends on what the network can provide. GHB needs secure summation among hospitals that do not collude; in exchange, its noise is calibrated to the whole network, it uses the hospitals' real case mix, and it needs no public feature distribution. On real cancer networks it was the most accurate method that a consortium could actually deploy, and under strong privacy the most accurate even against tuning on the test data. GHB-lite needs nothing but honest-but-curious neighbours, and each of its releases is private on its own. It is the better choice when a public feature distribution is known to match the hospitals' patients and when the network is large enough, in hospitals and in patients per hospital, for per-hospital releases to

be informative; the simulations describe that case. Real networks of a few to a few dozen small hospitals did not meet either condition. A network should also check, before committing, that privacy leaves something to share: Equations (4) and (3), with the network's total size in place of each hospital's for GHB, give that check from public quantities, namely the sizes, the budget and a prior guess of $\tau^2$.

### 8.2 Limits

GHB protects patients against honest-but-curious hospitals and any observer of the messages, under the guarantee of Proposition 5.1; hospitals that collude to remove their noise shares weaken it, and GHB-lite, whose releases are private on their own, is the option where collusion cannot be ruled out. A hospital that lies about its contribution is not excluded by the protocol itself. In Simulation 12 such lies never pushed honest hospitals below local performance under GHB-lite, because a large lie inflates $\hat{\tau}^2$ and switches borrowing off. GHB has no per-hospital release in which to see a lie, so its defence is each hospital's own evidence, which moves the weight to non-borrowing components when the network conflicts with the hospital's data; stronger adversaries call for robust aggregation [Blanchard et al. 2017] or verifiable summation. The privacy guarantee is per pair of releases, so refits compose and must be accounted for [Bun and Steinke 2016; Mironov 2017]. The Gaussian analysis is asymptotic; for very small $n_k$ the normal approximation is poor, which is one reason the method integrates over components fitted by maximum a posteriori rather than relying on closed forms. GHB's network estimate is a single Newton step from the origin. It is accurate when the pooled model is close to it, as for the clinical heads studied here; heads far from the origin would need a second round of releases, which the development data did not favour.

*What the evidence does not show.* Several components were added after a version failed, and were chosen with knowledge of that failure. In the simulations, for GHB-lite: the half-Cauchy prior on $\tau$ and the use of the anchor (Simulation 7); keeping the anchor out of the releases (Simulation 11); and, after the first run of Simulation 15, the score release, integrating $\tau$ out and the conflict check. On real data: the evidence-weighted default prior, secure aggregation of per-hospital quantities, the four extensions of the third pre-registration, and finally the joint releases of GHB, selected among about fifteen variants on twelve development networks. The confirmation guards against overfitting that selection, but it is one confirmation, on five networks of one data source, with one outcome and an eight-coefficient head. Its margins under weak privacy are small, and a different set of networks could reverse them. The baselines were tuned as well as a consortium could tune them, and their oracle arm shows what more tuning would buy, but other methods, other grids or other ways of tuning could do better. The simulations test internal consistency under Equation (2) and its logistic analogue, against adversaries we chose, and they were built for GHB-lite. The evidence says nothing about heads with many more coefficients, about outcomes other than mortality, about larger networks, or about participants who are adversarial in ways we did not test. GHB answers the question posed in Section 3 only under explicit assumptions: Gaussian site effects in the theory, honest and non-colluding participants, and a known collaboration graph. Relaxing each of them defines most of the open problems that follow.

### 8.3 Open problems

(1) **Private estimation of heterogeneity.** The value of collaboration is governed by $\tau^2$ (Equation (3)), yet hospitals rarely know it before joining a consortium. GHB does not estimate it from the network at all: it integrates over a fixed prior, weighted by each hospital's own evidence. GHB-lite estimates it from per-hospital releases, which is why it is more accurate in simulation, but those releases were too noisy and too biased on real networks. A private estimator of $\tau^2$

from joint releases, for instance from the secure sum of a quadratic statistic, would combine the strengths of both, and would let a hospital predict its benefit before committing. Its difficulty is sensitivity: a quadratic statistic of a hospital's score depends on that hospital's size, and our development attempts were too noisy to help.

(2) **Topology design.** Since mixing depends on the spectral gap (Proposition 4.3), which agreements to sign is an optimization problem with legal and economic constraints. Choosing a few long-range links that maximize the gap is a concrete, under-studied question.

(3) **Small-site privacy.** DP noise on per-hospital releases scales as $n_j^{-2}$ (Equation (4)), and GHB removes that cost by releasing only network sums. The price is trust in the summation. Shuffling, or secure summation that tolerates dropouts and detects collusion, would keep the noise calibrated to the network with a weaker trust model.

(4) **Incentives and valuation.** Large hospitals gain little from hierarchical borrowing and contribute most. Data valuation [Ghorbani and Zou 2019] adapted to privatized, decentralized settings could support fair credit, and regulation such as the European Health Data Space [European Parliament and Council of the European Union 2025] will shape what is feasible.

(5) **Shift-aware borrowing.** Proposition 4.1 treats heterogeneity as random. In practice it has structure, through case mix, devices and time. Combining causal invariance [Arjovsky et al. 2019; Castro et al. 2020] with hierarchical borrowing could separate the part of $\tau^2$ that is transportable from the part that is site-specific.

(6) **Group fairness.** GHB shrinks every coefficient of a hospital's head toward the network, including those through which predictions depend on patient groups, so it can carry other hospitals' disparities into a small hospital's model or dilute its own. Simulation 18 finds both effects small for GHB-lite, which estimates $\tau$ per coefficient: a disparity on which hospitals agree is borrowed and estimated better, and one on which they disagree is barely borrowed, while the fairness ratios of decisions stay at those of the true models. Keeping the disparity coefficient local is a privacy-free safeguard when disparities are expected to differ. GHB does not, however, pursue parity of decisions, and doing so in this setting is open: fairness-aware aggregation such as FAIR-FATE [Salazar et al. 2023] relies on a server and on many rounds of updates, and under requirement (R1) every fairness statistic that leaves a hospital would be a further private release that GHB's two releases do not budget for.

(7) **Procurement and strata.** The posterior variances of GHB tell every hospital which patients would be most informative to label next, and nobody has studied the joint design of labeling across a consortium. Equation (3) also applies with strata in place of hospitals, which suggests borrowing aimed at small subgroups rather than small sites.

## 9 CONCLUSION

Networks of small hospitals that cannot share patient data and have no trusted centre are the default organizational form of healthcare, and they are poorly served by methods that assume a server, fit one model, or must be tuned on data that small hospitals do not have. Graph Hierarchical Borrowing gives each hospital almost the same number of virtual patients, pays the privacy cost once, in two sums over the whole network, and runs over whatever agreements exist. Each hospital turns the network's two releases into a leave-one-out prior with the privacy noise propagated, and integrates the unknown heterogeneity out against its own evidence, so that nothing needs tuning. On five cancer networks that no part of its development had seen, and under a pre-registered protocol, GHB was more accurate than every federated and personalized method tuned as a consortium could tune it, at strong and at weak privacy. Under strong privacy it was more accurate than each

of them even when tuned on the test data. The margins are small under weak privacy, the evidence comes from one source of real networks, and the lighter variant that the simulations favour failed on real data. Whether the result holds for other outcomes, larger heads and larger networks is the main question this paper leaves open.

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